College of Computing and Data Science

Final-Year Project Report

Multi-Stakeholder Ethical Decision-Making Frameworks for AI Systems

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Abstract

As Artificial Intelligence (AI) systems become increasingly integrated into critical societal domains, the need for robust, transparent, and multi-stakeholder ethical evaluation becomes paramount. This report presents the design and implementation of the Multi-stakeholder Ethical Decision Framework (MEDF), a software platform developed to address this challenge. MEDF provides a structured, reproducible, and auditable workflow for evaluating AI applications against prominent governance frameworks, including the EU AI Act Assessment List for Trustworthy AI (ALTAI), the NIST AI Risk Management Framework (RMF), and Singapore's Model AI Governance Framework (MGAF). The system operationalizes high-level ethical principles into quantifiable metrics across six unified dimensions: transparency and explainability, fairness and non-discrimination, safety and robustness, privacy and data governance, human agency and oversight, and accountability.

By enabling configurable stakeholder weightings, MEDF surfaces and quantifies disagreements in ethical priorities, employing multi-objective optimization techniques (NSGA-II) to identify Pareto-optimal consensus solutions. This work details the system's three-tier architecture, the algorithmic design of its scoring (TOPSIS, WSM) and conflict resolution engines, and its application to three real-world case studies: live facial recognition, automated hiring, and clinical diagnostics. The resulting platform serves as a decision-support tool for developers, regulators, and affected communities, fostering a more rigorous and inclusive approach to AI governance.

Index Terms: AI Governance, Algorithmic Fairness, Decision Support Systems, Ethical AI, Explainable AI, Multi-Criteria Decision Making, Stakeholder Analysis, TOPSIS.

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Chapter 1

Introduction

The rapid proliferation of Artificial Intelligence (AI) has ushered in a new era of technological capability, with systems now deployed in high-stakes domains ranging from criminal justice and healthcare to finance and autonomous transportation. While the potential benefits are immense, the opacity, complexity, and scale of these systems introduce significant ethical risks. Incidents of algorithmic bias, discriminatory outcomes, privacy violations, and lack of accountability have become increasingly common, eroding public trust and highlighting the urgent need for systematic governance and oversight [1], [2]. In response, governments, academic institutions, and industry bodies have developed a variety of ethical principles and regulatory frameworks. However, a significant gap persists between high-level principles and their practical implementation in the software development lifecycle [3]. Developers and organizations often lack the specific tools and methodologies to translate abstract ethical goals into concrete, measurable, and auditable engineering requirements.

This Final-Year Project addresses this critical gap by designing, implementing, and evaluating the Multi-stakeholder Ethical Decision Framework (MEDF). MEDF is a decision-support platform that provides a structured and reproducible workflow for evaluating the ethical dimensions of AI applications. It operationalizes abstract principles from leading governance frameworks, the EU AI Act Assessment List for Trustworthy AI (ALTAI) [4], the NIST AI Risk Management Framework (RMF) [5], and Singapore's Model AI Governance Framework (MGAF) [6], into a unified set of six quantifiable dimensions: transparency and explainability, fairness and non-discrimination, safety and robustness, privacy and data governance, human agency and oversight, and accountability.

This report details the complete lifecycle of the MEDF project, from initial problem definition to final implementation and evaluation. It begins by reviewing the background and motivation for ethical AI governance, followed by a comprehensive literature review of existing frameworks and decision-making models. The core of the report is dedicated to the design and architecture of the MEDF system, including its three-tier structure, the algorithmic foundations of its scoring and conflict-resolution engines (TOPSIS, WSM, NSGA-II), and the rationale behind critical engineering decisions. The practical utility of the framework is demonstrated through the in-depth analysis of three real-world case studies: the Metropolitan Police's use of live facial recognition, iTutorGroup's automated hiring system, and the deployment of DeepMind's Streams application in the UK's National Health Service (NHS). The report concludes with a discussion of the project's limitations, its contributions to the field, and potential avenues for future research. The final output is a functional, open-source software artifact that serves as both a practical tool for AI governance and a reproducible research platform for future work in computational ethics. As documented in Chapter 13, the current codebase retains some separation-of-concerns technical debt, including conflict analysis logic that remains concentrated in the router layer rather than an extracted standalone engine module.

Chapter 2

Background and Motivation

The discourse surrounding AI ethics is not new, but its urgency has intensified as AI systems have transitioned from theoretical research to impactful, real-world applications. The motivation for this project stems from three interconnected observations: the escalating scale of AI-driven societal harm, the inadequacy of purely principles-based governance, and the complex, multi-stakeholder nature of ethical decision-making.

Firstly, the potential for AI to cause significant, often unintended, harm is now well-documented. High-profile examples include biased facial recognition systems that exhibit higher error rates for women and people of color [7], [8], automated hiring tools that perpetuate historical gender and age biases [9], and opaque credit scoring algorithms that deny loans without clear justification [10]. These failures are not merely technical. They have profound social, economic, and human rights implications. They underscore the reality that technical performance metrics, such as accuracy, are insufficient for evaluating the overall acceptability of an AI system. A broader, more holistic evaluative lens is required, one that explicitly accounts for societal values like fairness, accountability, and transparency [11].

Secondly, while the proliferation of ethical AI principles from governments and corporations is a positive development, these high-level declarations often fail to provide actionable guidance for practitioners [12]. A principle such as "AI should be fair" is a necessary starting point, but it does not specify what fairness means in a given context, how it should be measured, or how it should be balanced against other competing values like accuracy or security. This "principles-to-practice" gap leaves developers without the tools to navigate complex ethical trade-offs during the design and implementation process [3], [13]. The result is often an ad-hoc, inconsistent, and unauditable approach to ethics, where critical decisions are made without a formal, defensible methodology.

Thirdly, ethical questions in AI are rarely, if ever, one-dimensional. Different stakeholders, including developers, business owners, regulators, and the communities affected by the AI system, hold different values, priorities, and tolerances for risk [14]. A developer might prioritize system robustness and security, a regulator might focus on legal compliance and accountability, and an affected community member might be most concerned with fairness and the right to redress. A decision that appears optimal from one perspective may be unacceptable from another. This inherent multi-stakeholder conflict is a central challenge in AI governance [15]. Ignoring it leads to solutions that fail to achieve broad legitimacy and may ultimately be rejected by society. Therefore, a functional ethical framework must not only evaluate AI systems but also provide a mechanism for surfacing, understanding, and negotiating these diverse stakeholder perspectives.

This project is motivated by the conviction that a computational approach can help bridge these gaps. By creating a structured, data-driven decision framework, MEDF aims to make ethical evaluation more rigorous, transparent, and inclusive. It seeks to empower AI development teams with a practical tool that translates abstract principles into concrete analysis, facilitates a more democratic dialogue about ethical trade-offs, and ultimately contributes to the creation of more responsible and trustworthy AI systems [16], [17].

Chapter 3

Problem Definition

The central problem this project addresses is the **operationalization gap** in AI ethics: the significant disconnect between high-level ethical principles and the low-level technical and procedural mechanisms required to enforce them in practice. While numerous frameworks define *what* constitutes ethical AI, they provide limited guidance on *how* to systematically evaluate, measure, and govern these principles within a software development lifecycle [3], [12]. This gap manifests in three primary sub-problems:

1. **Lack of Standardized, Quantifiable Metrics.** Ethical concepts such as fairness, transparency, and accountability are abstract and context-dependent. Without a standardized methodology for translating these concepts into quantifiable metrics, evaluations remain subjective, inconsistent, and difficult to compare across different projects or over time. Developers lack a clear "yardstick" to measure their progress or to demonstrate compliance to regulators and the public.
2. **Unmanaged Stakeholder Conflict.** Different stakeholders, including developers, business owners, regulators and the communities affected by the AI system, hold different values, priorities and risk tolerances [14]. Although Freeman's stakeholder theory [14] and Mitchell et al.'s stakeholder salience model [15] originate in general management, their core constructs (stakeholder identification, competing interests, and salience-based prioritization) map directly to AI governance, where developers, regulators and affected communities hold structurally analogous roles with divergent risk tolerances. A developer might prioritize system robustness and security, a regulator might focus on legal compliance and accountability, and an affected community member might be most concerned with fairness and the right to redress. A decision that appears optimal from one perspective may be unacceptable from another. This inherent multi-stakeholder conflict is a central challenge in AI governance [15].
3. **Absence of Auditable and Reproducible Workflows.** Ethical assessments are often conducted in an ad-hoc manner, with results recorded in static documents that lack traceability to the underlying data, assumptions, and decision logic. This makes it nearly impossible to reproduce an evaluation, audit a decision process, or track the evolution of a system's ethical posture over time. For AI governance to be effective, it requires a level of rigor and transparency comparable to other critical software engineering processes like testing and security [18].

Therefore, the problem statement for this project is as follows:

> ***"To design and implement a computational framework that enables organizations to systematically and reproducibly evaluate AI systems against multiple ethical governance standards, while explicitly modeling and managing the conflicting priorities of diverse stakeholders."***

To the best of the author's knowledge, no existing tool simultaneously evaluates AI systems against multiple governance frameworks, detects conflicts between stakeholder priorities, or produces stakeholder-weighted ethical scores. MEDF is designed to fill this gap.

To achieve this, the project sets the following objectives:

- To develop a unified, six-dimensional model of AI ethics that harmonizes principles from leading international governance frameworks.
- To implement a multi-criteria decision-making (MCDM) engine using established algorithms (TOPSIS, WSM) to generate quantifiable ethical scores.
- To design and implement a mechanism for capturing stakeholder priorities as configurable weightings.
- To develop a conflict detection module that identifies and quantifies disagreement between stakeholder groups.
- To implement a multi-objective optimization algorithm (NSGA-II) to generate Pareto-optimal consensus solutions that help mediate stakeholder conflicts.
- To validate the framework's utility and effectiveness through its application to a set of diverse, real-world AI case studies.
- To deliver a completely documented, open-source software artifact that is both a practical tool and a platform for future research.

Chapter 4

Literature Review

This chapter reviews the foundational literature that informs the design of the MEDF platform. The review is organized into three areas. Firstly, it examines the foundational work of Zhang et al. on the Fairness in Design (FID) framework, which forms the direct theoretical basis for this project. Secondly, it surveys the broader landscape of AI governance frameworks, focusing on the three selected for implementation in MEDF: the EU ALTAI, NIST AI RMF, and Singapore's MGAF. Finally, it delves into the technical literature on Multi-Criteria Decision-Making (MCDM) methods, which provide the algorithmic foundation for MEDF's scoring and analysis capabilities.

4.1 Fairness in Design

The primary theoretical inspiration for this project is the work of Zhang, Shu, and Yu on the Fairness in Design (FID) framework [1]. Their 2023 paper, "Fairness in Design: A Framework for Facilitating Ethical Artificial Intelligence Designs," identifies a critical gap in the AI development lifecycle: the lack of a structured methodology for design teams to brainstorm and surface potential fairness issues during the initial design stage. The authors argue that while many tools exist for post-hoc bias detection in models and data, few resources are available to guide the more foundational, upstream process of designing fairness-aware systems from the outset.

To address this, they propose the FID framework, a game-like methodology centered around a set of prompt cards. These cards are designed to facilitate discussions among team members, encouraging them to consider fairness from the perspectives of various stakeholders. The framework distills the complex field of algorithmic fairness into ten major principles, which serve as brainstorming guides. User studies conducted by the authors demonstrated that FID was effective in helping participants make more nuanced decisions about fairness and lowered the barrier to entry for teams without deep pre-existing knowledge in the field. A related work by Zhang [19] further elaborates on this, positioning the FID toolkit as a critical intervention to embed ethical considerations early in the AI pipeline. An earlier conference version [20] and a follow-up study on explainability [21] provide additional context for this line of research.

MEDF builds directly on the philosophy of the FID framework. It takes the core idea of structured, stakeholder-centric ethical deliberation and seeks to operationalize it within a quantitative, computational platform. While FID is a qualitative, manual tool for brainstorming, MEDF is a quantitative, automated tool for evaluation and analysis. It extends the concept by not only facilitating the identification of issues but also by measuring their severity, quantifying the conflicts between stakeholder views, and proposing mathematically optimized resolutions. In this sense, MEDF can be seen as a downstream complement to the FID methodology: where FID helps shape the initial design, MEDF helps evaluate and govern the resulting implementation.

4.2 AI Governance Frameworks

The MEDF platform integrates and harmonizes three prominent AI governance frameworks, chosen for their distinct geographical origins and complementary perspectives. A comprehensive survey by Jobin, Ienca, and Vayena [2] identified a global convergence around five ethical principles (transparency, justice and fairness, non-maleficence, responsibility, and privacy), while also highlighting substantive divergence in how these principles are interpreted and implemented across different jurisdictions. The OECD Recommendation on Artificial Intelligence [22] further codifies these principles at an intergovernmental level. This finding directly motivates MEDF's multi-framework approach.

4.2.1 EU Assessment List for Trustworthy AI (ALTAI)

The European Union has taken a strong, rights-based approach to AI regulation, culminating in the proposed AI Act [23]. The High-Level Expert Group on AI (AI HLEG) published the "Ethics Guidelines for Trustworthy AI," which outlines seven requirements for achieving Trustworthy AI: (1) Human Agency and Oversight, (2) Technical Robustness and Safety, (3) Privacy and Data Governance, (4) Transparency, (5) Diversity, Non-discrimination and Fairness, (6) Societal and Environmental Well-being, and (7) Accountability [24]. To make these requirements actionable, the AI HLEG developed the Assessment List for Trustworthy AI (ALTAI) [4]. ALTAI is a practical checklist intended to help AI developers and deployers self-assess whether the AI system they are building is trustworthy. It breaks down each of the seven requirements into a series of concrete questions and considerations. Smuha [25] provides a detailed analysis of the EU's approach to these guidelines. MEDF uses the structure of ALTAI as a primary source for its unified dimension model, mapping ALTAI's requirements directly to its own evaluation criteria.

4.2.2 NIST AI Risk Management Framework (AI RMF)

Developed by the U.S. National Institute of Standards and Technology, the AI Risk Management Framework (AI RMF) provides a voluntary, flexible, and sector-agnostic structure for managing the risks associated with AI systems [5]. The AI RMF is organized around four core functions: Govern, Map, Measure, and Manage. It is designed to be integrated into an organization's broader risk management processes. Unlike the EU's more prescriptive, rights-based approach, the NIST framework is more focused on providing a process for identifying, assessing, and responding to risks throughout the AI lifecycle. It emphasizes the importance of context, continuous monitoring, and creating a culture of risk management. MEDF incorporates the risk-centric perspective of the NIST AI RMF, particularly in its harm assessment module, which seeks to quantify the potential severity of ethical risks.

4.2.3 Singapore Model AI Governance Framework (MGAF)

Singapore's Model AI Governance Framework (MGAF), now in its second edition, offers a pragmatic, industry-focused perspective on AI governance [6]. It is designed to be a readily implementable guide for organizations that are developing or deploying AI. The framework is built on two guiding principles: (1) organizations should be accountable for the AI they deploy, and (2) decisions made by AI should be explainable, transparent, and fair. The MGAF provides detailed guidance on internal governance structures, risk management in the AI lifecycle, and measures for customer relationship management. Its practical, business-oriented approach provides a valuable complement to the rights-based focus of the EU and the risk-management process focus of NIST. MEDF's emphasis on configurable stakeholder weights and trade-off analysis aligns well with the MGAF's pragmatic recognition that ethical AI involves balancing different commercial and societal interests.

4.3 Multi-Criteria Decision-Making (MCDM) Methods

The algorithmic core of MEDF is built upon established Multi-Criteria Decision-Making (MCDM) methods. MCDM is a sub-discipline of operations research that deals with making decisions in the presence of multiple, often conflicting, criteria [26]. This is directly analogous to the problem of ethical AI evaluation, where a system must be judged against multiple competing values. The challenge is compounded by the conceptual complexity of the criteria themselves: fairness in sociotechnical systems resists simple formalization [27], [28], and transparency requires context-sensitive approaches to explainability [29].

4.3.1 TOPSIS

The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) is a widely used MCDM method [30]. The fundamental principle of TOPSIS is that the chosen alternative should have the shortest geometric distance from the ideal solution and the longest geometric distance from the anti-ideal solution. The "ideal solution" is a hypothetical alternative that has the best possible value for each criterion, while the "anti-ideal solution" has the worst possible value. TOPSIS is computationally efficient and its logic is relatively easy to understand. It normalizes the data for each criterion and then calculates a closeness coefficient for each alternative, allowing them to be ranked [26]. MEDF uses TOPSIS as its primary scoring method due to its robustness and its ability to produce a clear, scalar score that represents the overall ethical performance of an AI system. Comparative analyses have shown that TOPSIS performs favorably against alternative MCDM methods such as VIKOR in multi-attribute decision problems [31].

4.3.2 NSGA-II and Pareto Optimization

When stakeholders have conflicting priorities, there is often no single solution that is optimal for everyone. In such cases, the goal is to find the set of "Pareto-optimal" solutions. A solution is Pareto-optimal if it is impossible to improve one stakeholder's outcome without worsening another's. The set of all such solutions is known as the Pareto frontier. NSGA-II (Non-dominated Sorting Genetic Algorithm II) is a popular and effective multi-objective evolutionary algorithm used to find this Pareto frontier [32]. It uses concepts of non-dominated sorting and crowding distance to guide its search, allowing it to find a diverse set of high-quality solutions along the frontier. MEDF employs NSGA-II in its Pareto router to generate a set of consensus weight vectors that represent the best possible compromises between conflicting stakeholder priorities.

Chapter 5

Ethical AI Governance Frameworks

This chapter provides a detailed analysis of the three governance frameworks integrated into MEDF and describes the process by which their diverse principles are harmonized into a single, unified evaluation model.

5.1 Selected Frameworks

5.1.1 EU Assessment List for Trustworthy AI (ALTAI)

The ALTAI framework, developed by the European Commission's High-Level Expert Group on AI [4], translates the seven requirements of the Ethics Guidelines for Trustworthy AI [24] into a structured self-assessment checklist, as detailed in Section 4.2.1. In the MEDF implementation, the framework definition (`eu_altai.yaml`) contains 6 criteria entries mapped across the six unified dimensions, with a total of 22 requirements. Intrinsic weights are derived from the number of assessment questions associated with each dimension.

5.1.2 NIST AI Risk Management Framework (AI RMF)

The NIST AI RMF [5], developed by the U.S. National Institute of Standards and Technology, takes a process-oriented approach organized around four core functions (Govern, Map, Measure and Manage), as detailed in Section 4.2.2. In the MEDF implementation, the framework definition (`nist_ai_rmf.yaml`) contains 6 criteria entries mapped across the six unified dimensions, with a total of 19 subcategories, capturing the risk-management perspective that complements the rights-based focus of ALTAI.

5.1.3 Singapore Model AI Governance Framework (MGAF)

The Singapore Model AI Governance Framework (MGAF) [6] is structured around two guiding principles and four areas of governance, offering a pragmatic industry-focused perspective, as detailed in Section 4.2.3. In the MEDF implementation, the framework definition (`sg_mgaf.yaml`) contains 6 criteria entries mapped across the six unified dimensions, with a total of 20 principles. The MGAF's emphasis on accountability is reflected in a higher intrinsic weight on the accountability dimension.

5.2 Harmonization and the Six Unified Dimensions

A central contribution of the MEDF design is the harmonization of the three source frameworks into a single, coherent evaluation model based on six Unified Dimensions. These dimensions were derived through a systematic analysis of the overlapping themes and requirements across all three frameworks, informed by the broader AI ethics literature [2], [33]. The six Unified Dimensions are:

1. **Transparency and Explainability.** This dimension assesses the degree to which the AI system's operations, decisions, and underlying logic can be understood by relevant stakeholders. It encompasses concepts such as traceability, interpretability, and clear communication of the system's capabilities and limitations.
2. **Fairness and Non-discrimination.** This dimension evaluates whether the AI system produces equitable outcomes and avoids unjust bias or discrimination against individuals or groups based on protected characteristics such as race, gender, or age.
3. **Safety and Robustness.** This dimension measures the technical reliability of the AI system, including its resilience to errors, adversarial attacks, and unexpected inputs. It also considers the system's ability to operate safely within its intended environment and to fail gracefully when encountering conditions outside its design parameters.
4. **Privacy and Data Governance.** This dimension assesses the system's compliance with data protection principles, including data minimization, purpose limitation, consent management, and the security of personal data throughout its lifecycle.
5. **Human Agency and Oversight.** This dimension evaluates the mechanisms for human oversight, intervention, and ultimate authority over the system's decisions, a principle strongly emphasized in the EU framework.
6. **Accountability.** A cornerstone of all three frameworks, this dimension assesses the mechanisms for redress, responsibility, and auditability. It evaluates who is responsible when things go wrong and whether there are clear processes for challenging the system's outcomes.

By mapping the specific clauses, principles, and questions from each of the three source frameworks to these six Unified Dimensions, MEDF creates a comprehensive and cohesive evaluation model. This allows the platform to provide a holistic assessment that is grounded in established international standards, while still offering the flexibility to analyze ethical performance at a granular, dimensional level. The framework definitions, including their mapping to the unified dimensions, are stored as version-controlled YAML files within the repository, ensuring that the foundation of the evaluation logic is itself transparent and auditable.

Chapter 6

MEDF Framework Design

The development of MEDF follows the Design Science Research Methodology (DSRM) proposed by Peffers et al. [34], which provides a structured process for creating and evaluating IT artifacts. The six DSRM activities map directly onto the structure of this project: problem identification and motivation (Chapters 2–3), objective definition (Chapter 3), design and development (Chapters 6–9), demonstration (Chapter 10), evaluation (Chapter 11) and communication (this report). This framing positions MEDF not merely as a software engineering exercise, but as a research artifact whose design decisions are traceable to an identified problem and whose utility is evaluated against defined objectives.

The Multi-stakeholder Ethical Decision Framework (MEDF) is designed as a comprehensive, software-enabled methodology to bridge the gap between high-level ethical principles and practical AI evaluation. The framework is built upon a conceptual architecture that guides a user through a structured workflow, from initial setup to final analysis and reporting. This chapter outlines the core design principles, the main components of the framework, and the intended user workflow.

6.1 Core Design Principles

The design of MEDF is guided by four fundamental principles:

- **Modularity and Extensibility.** The framework is designed to be modular. Governance frameworks, stakeholder profiles, and even scoring algorithms are implemented as distinct components. This allows the system to be easily extended in the future. For example, a new governance framework can be added by creating a new YAML definition file, without altering the core application logic.
- **Reproducibility and Auditability.** Every evaluation run through MEDF is intended to be reproducible. Given the same set of inputs (AI system scores, stakeholder weights, and framework selections), the system will always produce the same output. This is critical for auditability and for building trust in the evaluation process. All evaluation requests and results are logged, creating a complete audit trail.
- **Stakeholder-Centricity.** The framework explicitly acknowledges that ethical evaluation is a multi-stakeholder problem. It is designed not to find a single

"correct" answer, but to surface and navigate the tensions between different value systems. The configurable weighting system is the primary mechanism for achieving this, allowing the platform to model the unique priorities of developers, regulators, affected communities, and other relevant groups.

- **Quantification and Decision Support.** The framework translates qualitative ethical principles into quantifiable metrics. While recognizing that not all ethical nuances can be captured in numbers, this quantitative approach provides a clear, data-driven basis for comparison and decision-making. The goal is not to replace human judgment, but to augment it with structured, analytical insights.

6.2 Framework Components

The MEDF workflow is organized around several conceptual components:

1. **AI System Profile.** This is the object of evaluation. It consists of a description of the AI system and, most importantly, a set of baseline scores across the six Unified Dimensions. These scores, on a 1–7 Likert scale, represent an initial assessment of the system's ethical performance, typically provided by the development team or an initial assessor.
2. **Governance Framework Registry.** This component contains the definitions of the ethical frameworks (ALTAI, NIST RMF, MGAF). Each framework is defined in a YAML file, which specifies its dimensions and, critically, the intrinsic weight or importance of each dimension derived from its structure (e.g., the number of clauses or requirements related to that dimension). This provides a framework-specific prior for weighting.
3. **Stakeholder Profiles.** These profiles represent the different parties interested in the AI system's evaluation. Each profile has an ID, a role (e.g., Developer, Regulator), and a set of weights that define that stakeholder's priorities across the six Unified Dimensions. These weights represent how important each dimension is to that particular stakeholder.
4. **Scoring Engine.** This is the core analytical component. It takes the AI system's scores, the selected frameworks, and the stakeholder profiles as input. It then computes an "effective weight" for each dimension by combining the stakeholder's priorities with the framework's intrinsic emphasis (using a product pooling method). It then uses an MCDM algorithm (like TOPSIS) to calculate a final ethical score.
5. **Conflict and Consensus Engine.** This component analyzes the results from the scoring engine. It uses statistical methods (like Spearman's rank correlation) to measure the degree of agreement or disagreement between different stakeholders. When significant conflicts are detected, it can invoke a multi-objective optimization

algorithm (NSGA-II) to find a set of Pareto-optimal consensus solutions, compromise weightings that represent the best possible trade-offs between the conflicting parties.

6.3 User Workflow

The intended workflow for an analyst using the MEDF platform is as follows:

1. **Define the Evaluation Context.** The analyst begins by selecting the AI system to be evaluated and providing its baseline dimension scores. They also select the governance frameworks against which the system should be assessed (e.g., ALTAI and NIST RMF).
2. **Select Stakeholders.** The analyst chooses the relevant stakeholder profiles for the evaluation. They can use the default profiles (Developer, Regulator, Affected Community) or create custom profiles with specific weightings.
3. **Run Evaluation.** The analyst executes the evaluation. The MEDF backend calculates the per-framework and overall ethical scores, showing how the system performs from the perspective of each stakeholder and in aggregate.
4. **Analyze Conflicts.** The analyst then moves to the conflict analysis view. The platform displays the level of agreement between each pair of stakeholders, highlighting the specific dimensions where their priorities diverge most significantly.
5. **Explore Consensus.** If conflicts are high, the analyst can run the Pareto optimization engine. The platform will generate a set of alternative "consensus" weightings that represent viable compromises. The analyst can explore these solutions to understand the nature of the trade-offs required to resolve the ethical disagreements.
6. **Export and Report.** The analyst can export all the results, scores, conflict matrices, and consensus solutions, into a structured data format for reporting and audit purposes. The Streamlit interface provides visualizations and tables that can be directly used in reports and presentations.

This structured workflow is illustrated in Figure 6.1, which shows the complete stakeholder decision flow from profile definition through conflict detection to consensus generation.

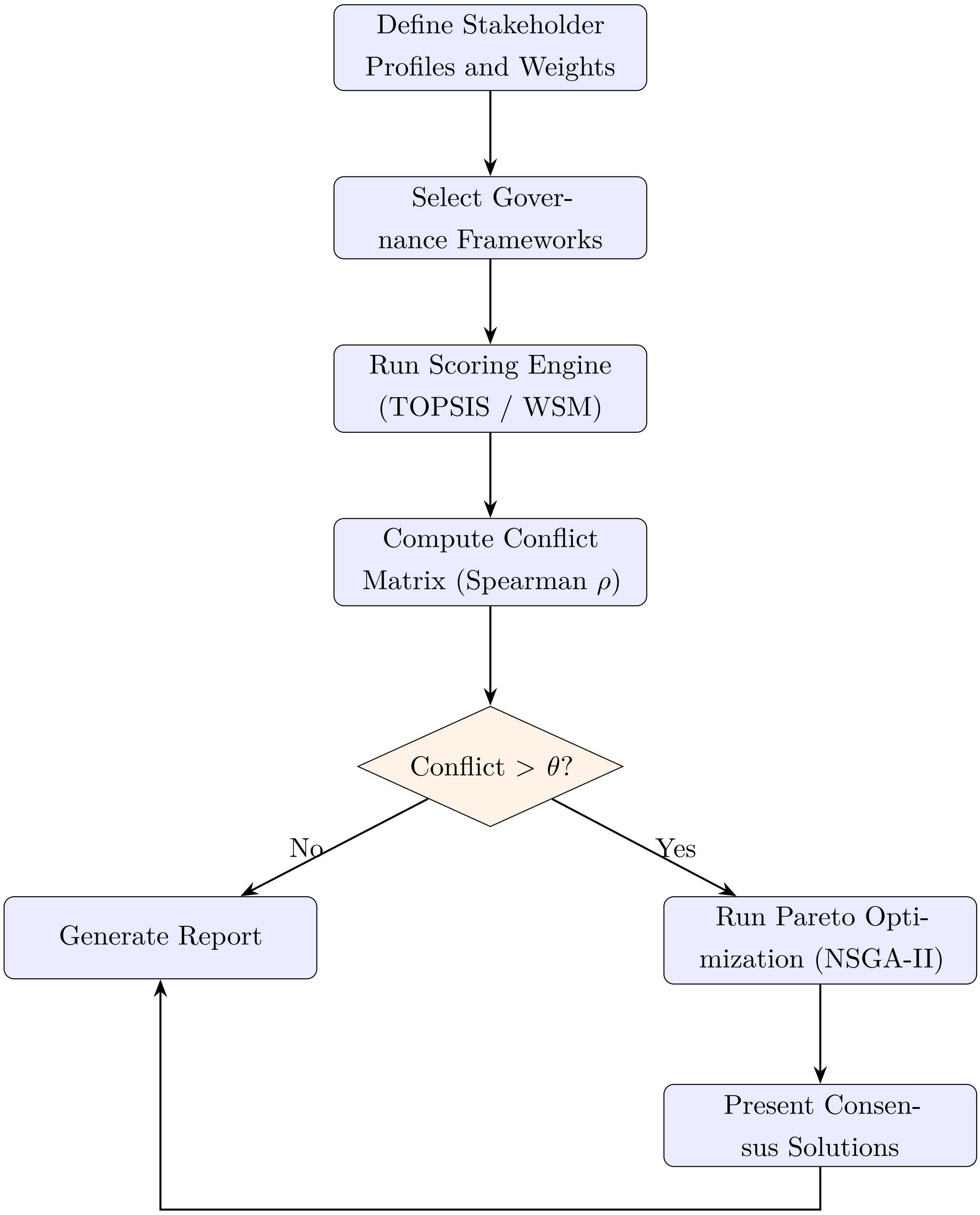

Figure 6.1: Stakeholder Decision Flow: value-weighted evaluation process.

This structured workflow ensures that the ethical evaluation is comprehensive, transparent, and grounded in a defensible methodology, transforming abstract ethical debates into a concrete, data-driven decision-making process.

Chapter 7

System Architecture

The MEDF platform is implemented as a modern web application with a clear separation of concerns, following a three-tier architecture. This design ensures modularity, scalability, and maintainability. The three primary layers are the Presentation Layer, the API/Orchestration Layer, and the Data/Configuration Layer. This chapter details the components within each layer and the interactions between them, as illustrated in Figure 7.1.

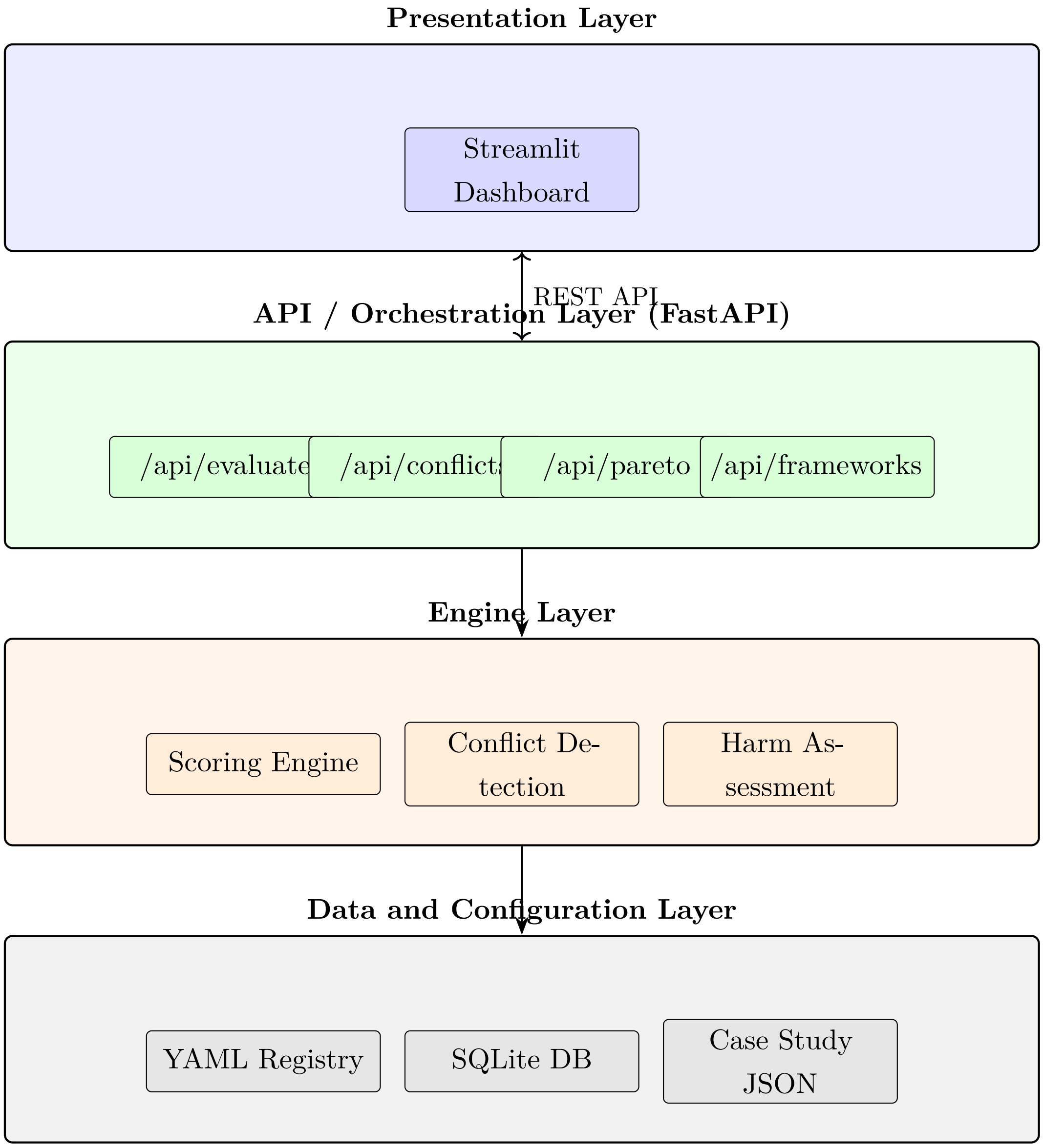

Figure 7.1: System Architecture: three-tier platform design.

7.1 Presentation Layer

The Presentation Layer is the user-facing component of the system, implemented as an interactive web dashboard using Streamlit. The complete frontend is contained within a single Python script, `streamlit_app.py`. This choice was made to facilitate rapid prototyping and tight integration with the Python-based data analysis and visualization libraries used to display evaluation results.

The dashboard provides a graphical user interface (GUI) for analysts to:

- Configure and initiate evaluation runs.
- Define or select AI systems and their baseline dimension scores.
- Select and customize stakeholder profiles and their priority weightings.
- View evaluation results, including overall scores, per-framework breakdowns, and dimensional contributions.
- Analyze stakeholder conflicts through correlation matrices and divergence tables.
- Trigger and explore Pareto-optimal consensus solutions.
- Export evaluation data and visualizations for reporting.

The Streamlit application communicates with the backend exclusively through the defined REST API, ensuring a clean separation between the user interface and the core business logic.

7.2 API/Orchestration Layer

The backend is a robust API server built with FastAPI. FastAPI was chosen for its high performance, native support for asynchronous operations, and automatic generation of interactive OpenAPI documentation, which provides excellent visibility into the API contract for development and auditing. The API layer is responsible for orchestrating the complete evaluation workflow.

It is composed of several modular routers, each handling a specific domain of functionality:

- `/api/frameworks`: Serves the definitions of the available governance frameworks.
- `/api/stakeholders`: Manages the creation, retrieval, and updating of stakeholder profiles.
- `/api/evaluate`: The primary endpoint for running an ethical evaluation. It receives the evaluation context and returns the computed scores.
- `/api/conflicts`: Analyzes the evaluation results to identify and quantify disagreements between stakeholders.

- `/api/pareto`: Executes the multi-objective optimization to find consensus solutions when conflicts are detected.

This layer acts as the central nervous system of the application, validating all incoming requests against strict Pydantic data models (`app/models.py`) and coordinating the execution of tasks by the Engine Layer.

7.3 Engine Layer

This layer contains the core algorithmic and business logic of the MEDF platform. It consists of several specialized Python modules:

- **Scoring Engine** (`scoring_engine.py`): Implements the MCDM algorithms, including TOPSIS and WSM, for calculating ethical scores.
- **Conflict Analysis** (`routers/conflicts.py`): Implements the full stakeholder conflict analysis workflow, including Spearman's rank correlation, divergent-dimension identification, conflict severity determination, and integration with the harm-assessment module. In the current codebase, this logic remains within the router layer rather than a dedicated engine module, which is a maintainability limitation rather than a missing functional capability.
- **Harm Assessment** (`harm_assessment.py`): Implements a model to estimate the potential severity of ethical risks. The harm score is computed as a weighted combination: $h = 0.7 \times (1 - s_{\text{norm}}) + 0.3 \times \bar{d}$, where s_{norm} is the normalized dimension score and $\bar{d}$ is the mean pairwise stakeholder weight disagreement.
- **Pareto Engine** (`routers/pareto.py`): Integrates the NSGA-II algorithm via the `pymoo` library to compute the Pareto frontier of consensus weightings. A random-sampling fallback is provided if `pymoo` is unavailable.

These engines are stateless and operate on the data passed to them by the API layer, ensuring that the core logic is decoupled from the web-serving and data persistence concerns.

7.4 Data and Configuration Layer

This layer is responsible for all data persistence and static configuration. It comprises:

- **Framework YAML Registry.** The definitions for ALTAI, NIST RMF, and MGAF are stored in human-readable YAML files. This allows the governance criteria to be version-controlled and updated without changing application code.

The `framework_registry.py` module is responsible for loading and validating these files at startup.

- **SQLite Database.** A lightweight SQLite database is used to store stakeholder profiles. SQLAlchemy, a Python ORM, is used to interact with the database, providing an abstraction layer that would facilitate migration to a more powerful database system like PostgreSQL if needed.
- **Case Study JSON Files.** The three case studies are defined in structured JSON files, containing their baseline scores and contextual information. This allows for consistent and reproducible testing and demonstration.
- **Audit Logs.** To ensure complete traceability, the system can be configured to write detailed JSONL audit logs for every API request and response, capturing the exact inputs and outputs of every evaluation.

This layered architecture, summarized in the module diagram in Figure 7.2, creates a robust and maintainable system where each component has a clearly defined responsibility.

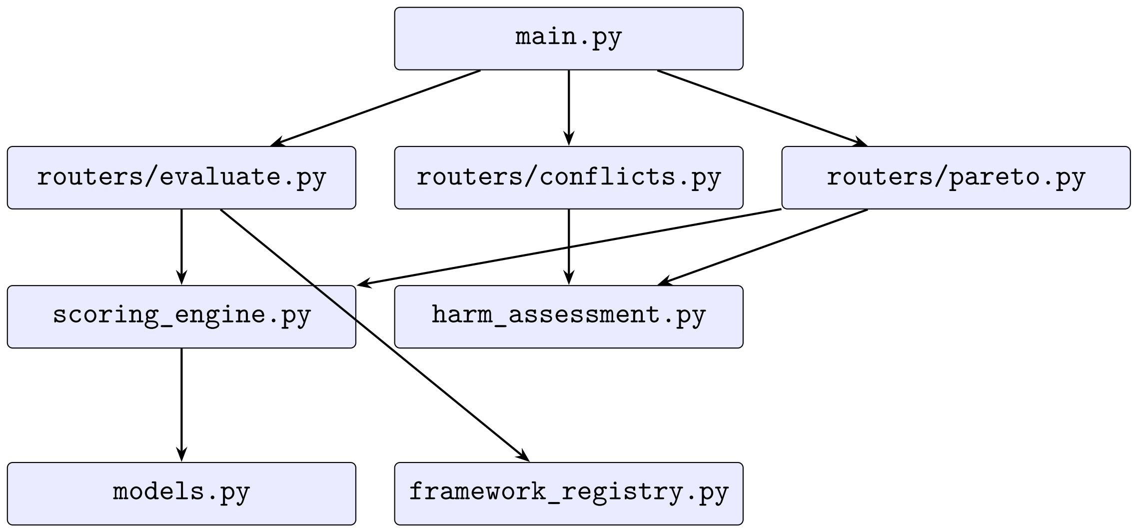

Figure 7.2: Software Module Diagram: component dependencies and data flow.

Chapter 8

Algorithm Design and Analysis

The analytical power of the MEDF platform is derived from a set of carefully selected and implemented algorithms that translate qualitative ethical considerations into a quantitative, decision-oriented format. This chapter details the design of the core algorithms responsible for scoring, weighting, and conflict resolution. All core modeling decisions, experimental design, and final validation were performed by the author.

8.1 Ethical Scoring Engine

The scoring engine is the heart of the evaluation pipeline. Its purpose is to compute a single, normalized score representing the ethical performance of an AI system, taking into account the system's characteristics, the chosen governance framework, and the priorities of a given stakeholder. This process involves three steps: weighting, aggregation, and normalization.

8.1.1 Stakeholder and Framework Weighting

The evaluation process begins by establishing the "effective weights" for each of the six Unified Dimensions. This is not simply the stakeholder's stated priorities. Instead, MEDF computes a combined weight that reflects both the stakeholder's values and the intrinsic emphasis of the selected governance framework. This is achieved through a **product pooling method**:

1. **Stakeholder Weights** ($\mathbf{w}_s$): Each stakeholder provides a vector of weights, where each element represents the importance they assign to a dimension. This vector is normalized to sum to 1.0.
2. **Framework Weights** ($\mathbf{w}_f$): For each governance framework, a vector of intrinsic weights is pre-calculated. This is derived from the structure of the framework itself, for example, by counting the number of requirements, clauses, or principles related to each dimension. This vector is also normalized to sum to 1.0.
3. **Effective Weights** ($\mathbf{w}_{\text{eff}}$): The effective weight for each dimension is calculated as the element-wise product of the stakeholder and framework weight vectors. The resulting vector is then re-normalized to sum to 1.0:

$$\mathbf{w}_{\text{eff}} \propto \mathbf{w}_s \odot \mathbf{w}_f \tag{8.1}$$

This product pooling approach ensures that a dimension receives a high effective weight only if *both* the stakeholder and the framework consider it important. A dimension that is critical to the stakeholder but not emphasized by the framework (or vice versa) will receive a moderate effective weight. This prevents any single source of weighting from dominating the evaluation and produces a more balanced, context-sensitive result.

8.1.2 TOPSIS for Score Aggregation

TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) [26], [30] is the primary scoring method in MEDF. Given a decision matrix $\mathbf{D}$ containing the AI system's scores across the six dimensions and the effective weight vector $\mathbf{w}_{\text{eff}}$, the TOPSIS algorithm proceeds as follows:

Step 1: Normalize the Decision Matrix. Raw Likert scores (1–7) are scaled to the $[0, 1]$ range:

$$x'_{ij} = \frac{x_{ij} - 1}{7 - 1} \tag{8.2}$$

Note on single-alternative evaluation. Standard TOPSIS is designed to rank multiple alternatives. When MEDF evaluates a single AI system, the implementation constructs a three-row decision matrix by augmenting the real score vector with two synthetic rows: one representing the theoretical best case (all dimensions set to the maximum Likert score) and one representing the theoretical worst case (all dimensions set to the minimum). The ideal and anti-ideal solutions are then derived from the column maxima and minima of the weighted normalized matrix in the standard manner (Step 4 below). This synthetic-anchor approach ensures that the closeness coefficient remains well-defined and interpretable as a distance-to-ideal metric, even in the absence of multiple real alternatives for comparison. The real system's score is extracted from row 0 of the resulting coefficient vector.

Step 2: Vector Normalization. The scaled matrix is then vector-normalized:

$$r_{ij} = \frac{x'_{ij}}{\sqrt{\sum_i (x'_{ij})^2}} \tag{8.3}$$

Step 3: Weighted Normalized Matrix. The normalized values are multiplied by the effective weights:

$$v_{ij} = r_{ij} \times w_{\text{eff},j} \tag{8.4}$$

Step 4: Determine Ideal and Anti-Ideal Solutions. For benefit criteria (where higher is better), the ideal solution A^+ contains the maximum values and the anti-ideal solution A^- contains the minimum values:

$$A^+ = \{\max(v_{ij})\}, \quad A^- = \{\min(v_{ij})\} \tag{8.5}$$

Step 5: Calculate Euclidean Distances. The distance of each alternative to the ideal and anti-ideal solutions is computed:

$$d^+ = \sqrt{\sum_j (v_{ij} - A_j^+)^2}, \quad d^- = \sqrt{\sum_j (v_{ij} - A_j^-)^2} \tag{8.6}$$

Step 6: Closeness Coefficient. The final TOPSIS score is the closeness coefficient:

$$C = \frac{d^-}{d^+ + d^-} \tag{8.7}$$

A score of $C = 1.0$ indicates a perfect match with the ideal solution, while $C = 0.0$ indicates a match with the anti-ideal solution. The scoring aggregation process is illustrated in Figure 8.1.

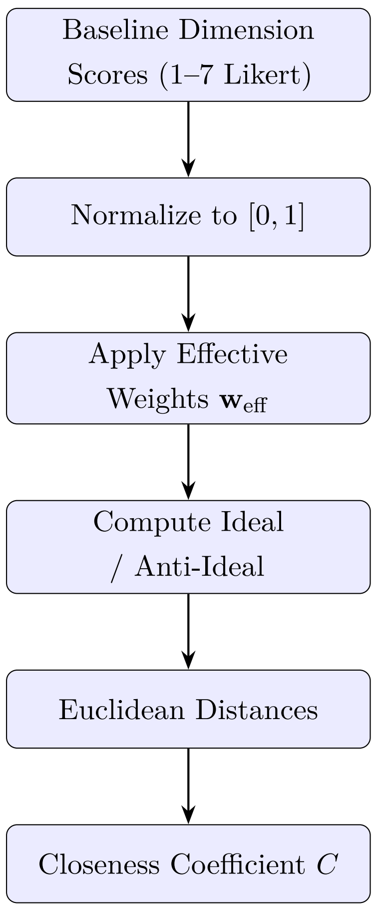

Figure 8.1: Scoring Aggregation: multi-framework evaluation pipeline.

8.2 Conflict Detection and Consensus Generation

8.2.1 Conflict Detection with Spearman's Correlation

To quantify the degree of agreement or disagreement between stakeholder groups, MEDF uses Spearman's rank correlation coefficient (ρ). For each pair of stakeholders, the algorithm computes the correlation between their weight vectors. A ρ close to $+1$ indicates strong agreement (both stakeholders rank the dimensions in a similar order of importance), while a ρ close to -1 indicates strong disagreement (their priorities are inversely related). A ρ near 0 indicates no significant relationship.

The conflict detection module computes two types of correlation matrices:

1. **Priority Conflict (Weights-Only).** This matrix computes Spearman's ρ directly between the raw stakeholder weight vectors. It measures whether stakeholders agree on which dimensions are most important, independent of the AI system being evaluated.
2. **System-Salience Conflict (Contribution-Based).** This matrix computes Spearman's ρ between the "contribution vectors," which are the element-wise products of the effective weights and the system's dimension scores. This measures whether stakeholders agree on which dimensions contribute most to the final score for a specific AI system. Two stakeholders might disagree on general priorities but agree on the critical issues for a particular system.

8.2.2 Pareto Optimization with NSGA-II

When the conflict analysis reveals significant disagreement between stakeholders, MEDF employs the NSGA-II algorithm [32] to find a set of Pareto-optimal consensus weight vectors. The optimization problem is formulated as follows:

For each stakeholder k, define an objective function that measures the weighted distance between a candidate consensus weight vector $\mathbf{w}_c$ and the stakeholder's preferred weight vector $\mathbf{w}_k$:

$$f_k(\mathbf{w}_c) = \sum_i s_i \cdot |w_{c,i} - w_{k,i}| \tag{8.8}$$

where s_i is a salience factor derived from the AI system's dimension scores, ensuring that the optimization prioritizes agreement on the dimensions that matter most for the system under evaluation.

NSGA-II then searches for the set of weight vectors that minimize all objective functions simultaneously. The resulting Pareto frontier represents the set of all possible compromises where no stakeholder can be made better off without making another worse off.

Each point on the frontier is a viable consensus solution, and the decision-makers can select the one that best reflects the desired balance of interests.

8.3 Weighted Sum Model (WSM)

In addition to TOPSIS, MEDF implements the Weighted Sum Model (WSM) as an alternative scoring method. WSM computes the overall score as a simple weighted average of the normalized dimension scores:

$$S_{\text{WSM}} = \sum_j w_{\text{eff},j} \times x'_j \tag{8.9}$$

WSM is simpler and more intuitive than TOPSIS but does not account for the distance to ideal and anti-ideal solutions. It is provided as an option for users who prefer a more straightforward aggregation method or who wish to compare results across different scoring approaches.

8.4 Harm Assessment Model

The harm assessment module provides a high-level risk classification based on the evaluation results. Two complementary threshold systems are used, reflecting the different scales of the underlying metrics.

The **evaluation router** classifies the TOPSIS closeness coefficient C (where higher values indicate better ethical performance) using the thresholds in Table 8.1.

Table 8.1: Risk Classification: evaluation thresholds based on closeness coefficient C.

Score Range	Risk Level	Description
$C \geq 0.80$	Low	Acceptable ethical performance.
$0.60 \leq C < 0.80$	Medium	Moderate ethical issues.
$0.40 \leq C < 0.60$	High	Significant ethical concerns.
$C < 0.40$	Critical	Severe ethical deficiencies.

The **harm assessment module** classifies a composite harm score h (where higher values indicate greater harm) using the thresholds in Table 8.2. The harm score is computed as $h = 0.7 \times (1 - s_{\text{norm}}) + 0.3 \times \bar{d}$, where s_{norm} is the normalized dimension score and $\bar{d}$ is the mean pairwise stakeholder weight disagreement. The 0.7/0.3 weighting reflects a deliberate design decision: a system's own ethical performance should dominate its risk classification, with stakeholder disagreement serving as a secondary aggravating factor that elevates the harm estimate when governance consensus is lacking.

Table 8.2: Harm Severity Classification: thresholds based on harm score h.

Harm Score Range	**Severity**	**Description**
$h < 0.25$	Low	Minimal ethical risk.
$0.25 \leq h < 0.50$	Moderate	Notable ethical risk.
$0.50 \leq h < 0.75$	High	Serious ethical risk.
$h \geq 0.75$	Critical	Severe ethical risk.

These two systems are conceptually consistent: the closeness coefficient is a “higher is better” metric while the harm score is a “higher is worse” metric. A system with a low closeness coefficient will tend to have a high harm score, and vice versa. The harm assessment also considers the level of stakeholder conflict. A system with a moderate overall score but high stakeholder conflict may be flagged as higher risk than its score alone would suggest, reflecting the additional governance challenge posed by unresolved ethical disagreements.

8.5 End-to-End Evaluation Pipeline

The complete MEDF evaluation pipeline integrates all the components described above into a coherent, sequential workflow. Figure 8.2 illustrates the end-to-end process from input to output.

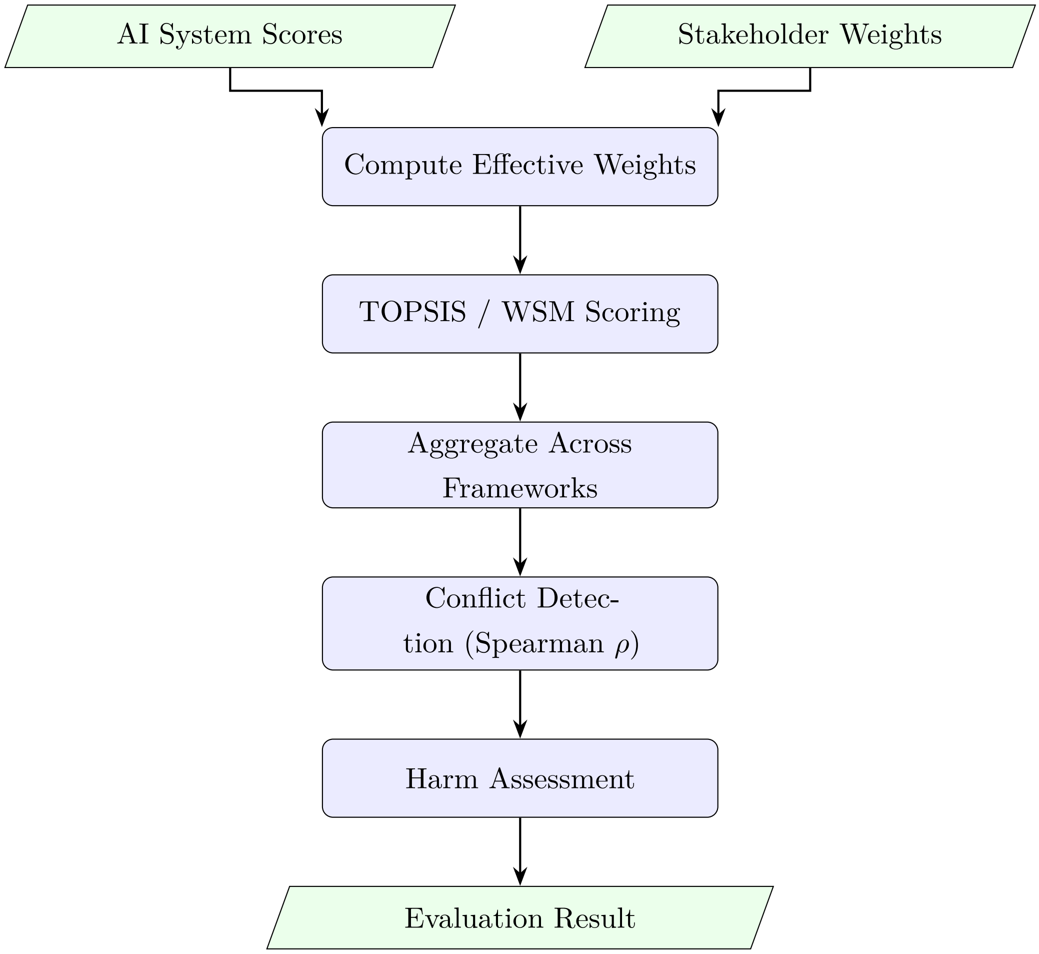

Figure 8.2: End-to-End Evaluation Pipeline: scoring, aggregation, and harm assessment stages.

Chapter 9

Implementation

This chapter details the technical implementation of the MEDF platform, translating the architectural and algorithmic design into a concrete software artifact. The implementation is based on a modern Python technology stack and emphasizes code quality, testability, and reproducibility. The complete source code is available in the accompanying GitHub repository.

9.1 Technology Stack

The project uses a curated set of open-source libraries and frameworks:

- **Backend Framework.** FastAPI is used for the API server, providing a high-performance, asynchronous foundation with excellent data validation capabilities through its integration with Pydantic.
- **Frontend Framework.** Streamlit is used for the interactive web dashboard, enabling rapid development of data-centric user interfaces directly in Python.
- **Data Persistence.** SQLAlchemy serves as the Object-Relational Mapper (ORM) for interacting with a simple SQLite database. This choice prioritizes ease of setup and reproducibility for a research prototype, while the ORM provides a clear path for future migration to a more robust database.
- **Numerical and Scientific Computing.** NumPy is used for all numerical operations, particularly in the scoring and conflict detection engines, providing efficient and reliable array computations. The `pymoo` library is used for the implementation of the NSGA-II algorithm.
- **Configuration.** Framework definitions are managed using YAML, a human-readable data serialization format, which allows for easy editing and versioning of governance criteria.
- **Testing.** Pytest is used as the testing framework, with a comprehensive suite of 105 tests across 33 test files covering unit, integration, property-based, and end-to-end scenarios.

9.2 Codebase Structure and Critical Modules

The repository is organized into several distinct directories, reflecting the modular architecture. The core application logic resides in the `app/` directory.

9.2.1 `app/main.py`

This is the main entry point for the FastAPI application. It is responsible for initializing the application, setting up the database connection, loading the framework definitions from the YAML registry at startup, and including the API routers.

9.2.2 `app/models.py`

This module contains all the Pydantic data models that define the API schemas for requests and responses. It also includes the SQLAlchemy ORM models for the database tables (e.g., `DBStakeholderProfile`). This strict, centralized schema definition is critical for ensuring data consistency and providing clear API contracts.

9.2.3 `app/scoring_engine.py`

This module contains the core algorithmic implementation for the scoring process. It includes functions for Likert score normalization, weight validation, and the implementations of the TOPSIS and WSM scoring methods. The code is written using NumPy for efficient vector and matrix operations.

9.2.4 `app/routers/`

The `app/routers/` directory contains the individual FastAPI routers for each major API endpoint:

- `evaluate.py` handles the `/api/evaluate` endpoint. It receives the `EvaluateRequest` payload, computes effective weights through section-based product pooling of stakeholder and framework weights, coordinates calls to the scoring engine, and returns a validated `EvaluationResult` response.
- `conflicts.py` handles the `/api/conflicts` endpoint and is the *de facto* conflict analysis engine. It implements the complete Spearman rank correlation computation for both weights-only and contribution-based conflict metrics, identifies divergent dimensions, determines conflict severity levels, and integrates the harm assessment

module. As noted in Chapter 7, this logic currently remains in the router layer rather than an extracted standalone engine module.

- `pareto.py` handles the `/api/pareto` endpoint and implements the NSGA-II multi-objective optimization for computing Pareto-optimal consensus weight vectors.

This modular approach keeps the endpoint logic clean and organized, though the concentration of conflict analysis logic within the router rather than a dedicated engine module is an acknowledged technical debt item.

9.3 Design Decisions and Implementation Rationale

A critical aspect of the implementation process was making deliberate engineering decisions to balance rigor, reproducibility, and development velocity. These decisions are documented in `docs/design_decisions.md`.

- **Why Weighted Scoring?** A simple unweighted average of dimension scores would fail to capture the differential importance of ethical criteria. Weighted scoring, where weights are explicitly defined by stakeholders and frameworks, provides a more nuanced and realistic evaluation model.
- **Why Configurable Stakeholder Weights?** Ethical priorities are subjective and context-dependent. Hardcoding a single set of weights would impose a single value system. Making weights configurable is the core mechanism that allows MEDF to model multi-stakeholder perspectives and surface conflicts.
- **Why a Sequential Evaluation Pipeline?** The evaluation pipeline (weighting, scoring, aggregation) is designed as a deterministic, sequential process. This ensures that the results are reproducible and the logic is easy to follow and audit. Parallelizing these steps would introduce unnecessary complexity without a clear benefit for this application.
- **Why a Three-Tier Architecture?** The separation of the presentation (Streamlit), API (FastAPI), and data layers creates a loosely coupled system. This allows each component to be developed, tested, and scaled independently. For example, the Streamlit frontend can be replaced with a React application without requiring any changes to the backend API.

9.4 Reproducibility and Testing

Ensuring the reproducibility of the research artifact was a primary implementation goal. This was achieved through several mechanisms:

- **Dependency Pinning.** All Python dependencies are pinned to specific versions in `requirements.txt` to ensure a consistent runtime environment.
- **Deterministic Algorithms.** The core scoring algorithms are deterministic. The NSGA-II implementation uses a fixed random seed during evaluation runs to ensure that the Pareto frontier results are also reproducible.
- **Comprehensive Test Suite.** The Pytest suite includes tests that lock down the API contract (`test_api_contract_lock.py`), verify the mathematical correctness of the scoring engine against hand-calculated examples (`test_topsis_hand_verification.py`), and run end-to-end evaluations with the case study data to prevent regressions.
- **Continuous Integration.** GitHub Actions are configured to run the complete test suite on every push and pull request, ensuring that the codebase remains in a valid and tested state at all times.

This rigorous approach to implementation and testing ensures that MEDF is not just a conceptual framework, but a reliable and defensible software tool.

The MEDF platform is deployed and publicly accessible at `https://ccds25-0582-medf-api.streamlit.app`, active as of the submission date (April 17, 2026). Local setup instructions are provided in the repository `README.md`, enabling full reproduction of all results presented in this report.

Chapter 10

Platform Interface

The MEDF platform is equipped with an interactive web-based dashboard built using Streamlit. This interface provides a user-friendly environment for analysts, developers, and other stakeholders to conduct ethical evaluations without needing to interact directly with the backend API or the codebase. The dashboard is designed to guide the user through the evaluation workflow, from configuration to results analysis. This chapter describes the critical features and components of the user interface, illustrated with screenshots captured from the live application running on localhost. All screenshots in this chapter were captured from the live deployment described in Section 9.4.

10.1 Main Evaluation View

The primary view lets the user configure and run a new evaluation by selecting an AI system, setting baseline dimension scores on the 1–7 Likert scale and choosing the frameworks and stakeholder profiles to include in the analysis. A "Run Demo Scenario" button populates the form with pre-configured case study data to support quick demonstration and testing. Figure 10.1 shows the evaluation configuration interface.

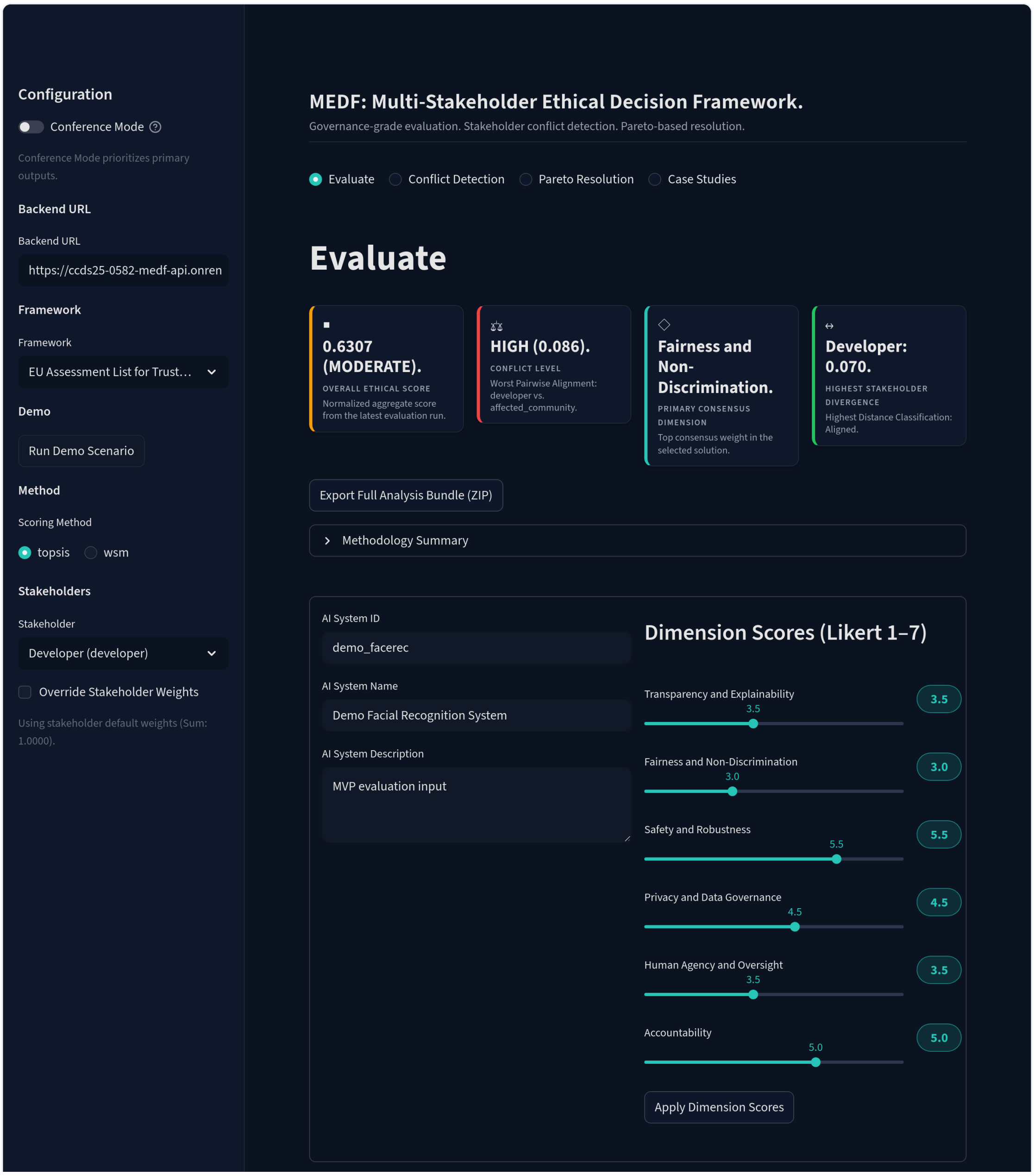

Figure 10.1: Evaluation Configuration: dimension score sliders, framework selection, and stakeholder weight customization.

10.2 Results Visualization

Once an evaluation completes, the dashboard presents the outcome as an overall score gauge (color-coded by risk level), a framework score breakdown bar chart and a dimensional contribution radar chart that surfaces the ethical strengths and weaknesses of the system. Figure 10.2 shows the evaluation results after running the demo scenario.

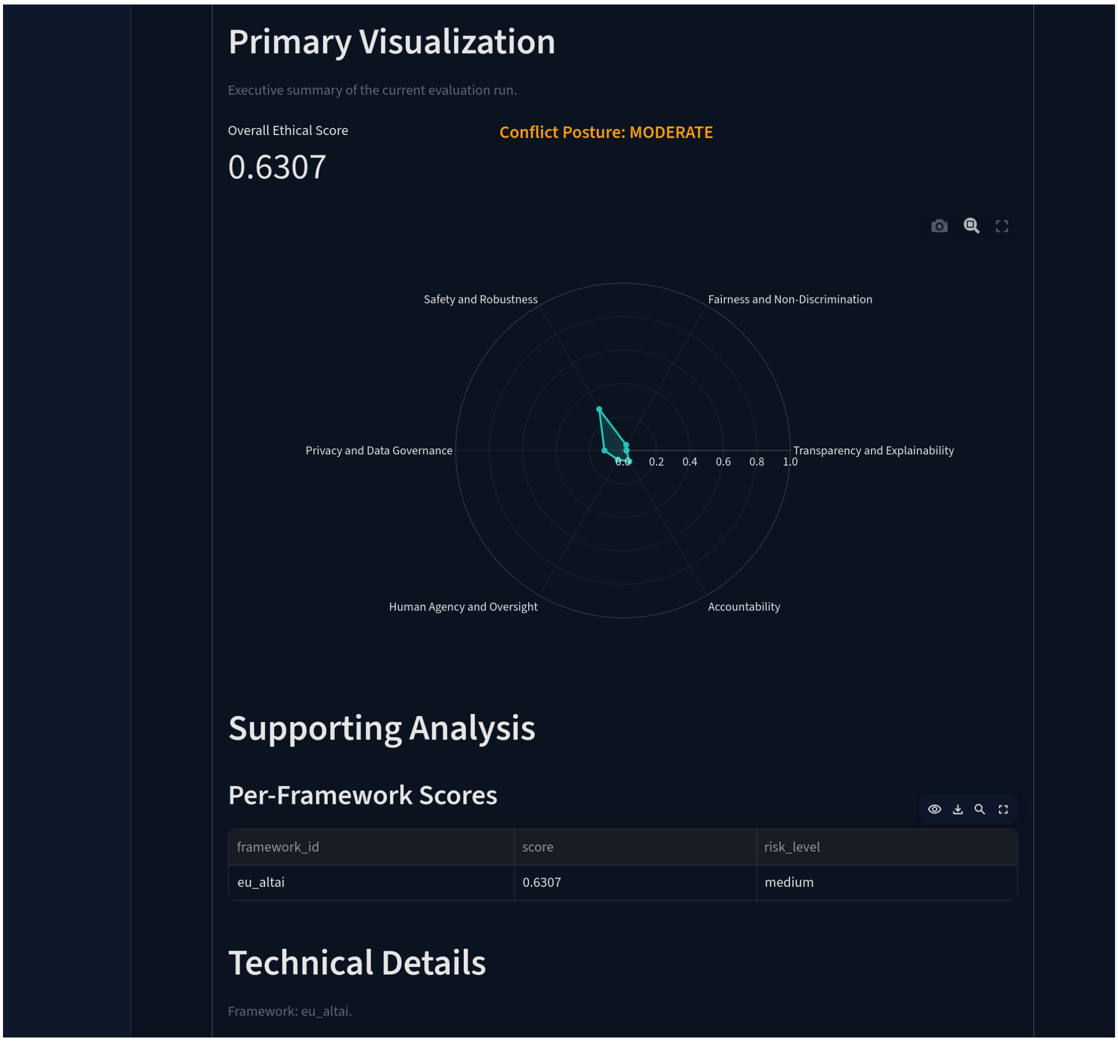

Figure 10.2: Evaluation Results: overall score, framework breakdown, and dimensional contributions.

10.3 Conflict and Consensus Analysis

The conflict analysis view displays a heatmap of Spearman's rank correlation coefficients between stakeholder pairs and, when Pareto optimization is run, a plot of non-dominated consensus solutions that lets the user visually explore trade-offs between compromise options. Figure 10.3 shows the conflict detection interface and Figure 10.4 shows the Pareto resolution interface.

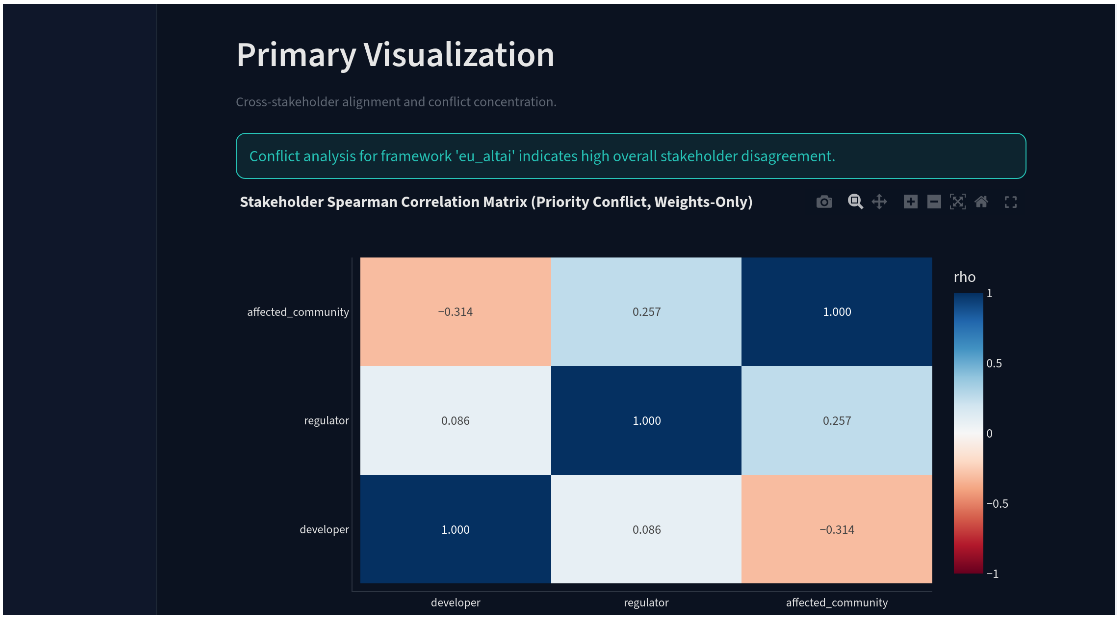

Figure 10.3: Conflict Detection: stakeholder conflict matrix heatmap.

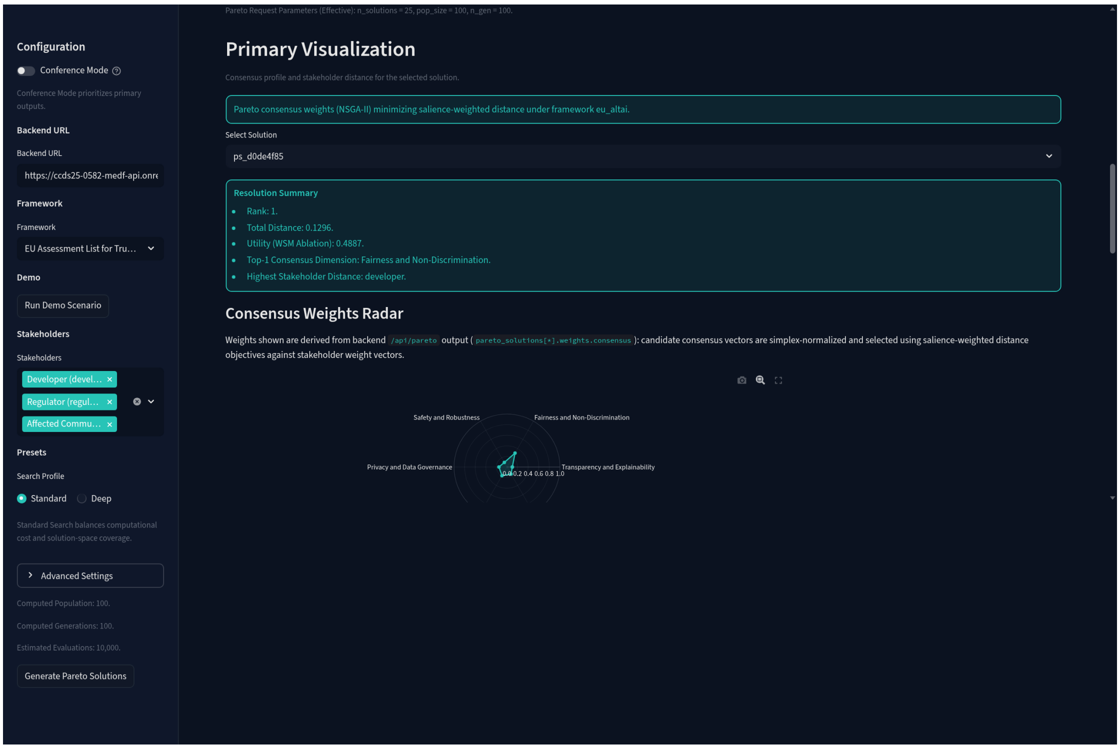

Figure 10.4: Pareto Resolution: consensus optimization results.

10.4 Case Study Browser

The Case Studies tab lets the user browse and run the three pre-configured case studies (Facial Recognition, Hiring Algorithm, Healthcare Diagnostic), each displayed with a system summary, deployment context, known risks and evidence sources, with a single click triggering a complete evaluation to reproduce the results in Chapter 11. Figure 10.5 shows the case study browser, and Figure 10.6 shows the results after running Case Study 1.

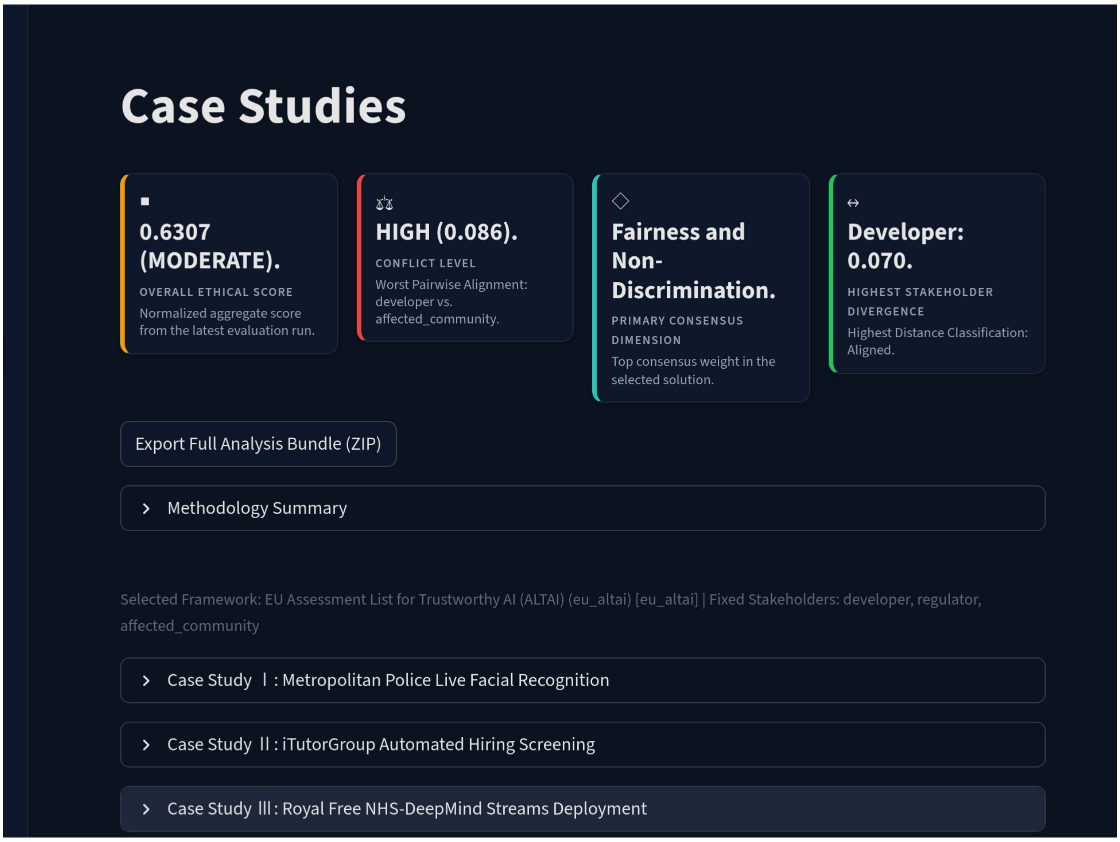

Figure 10.5: Case Study Browser: pre-configured case studies with system descriptions and evidence sources.

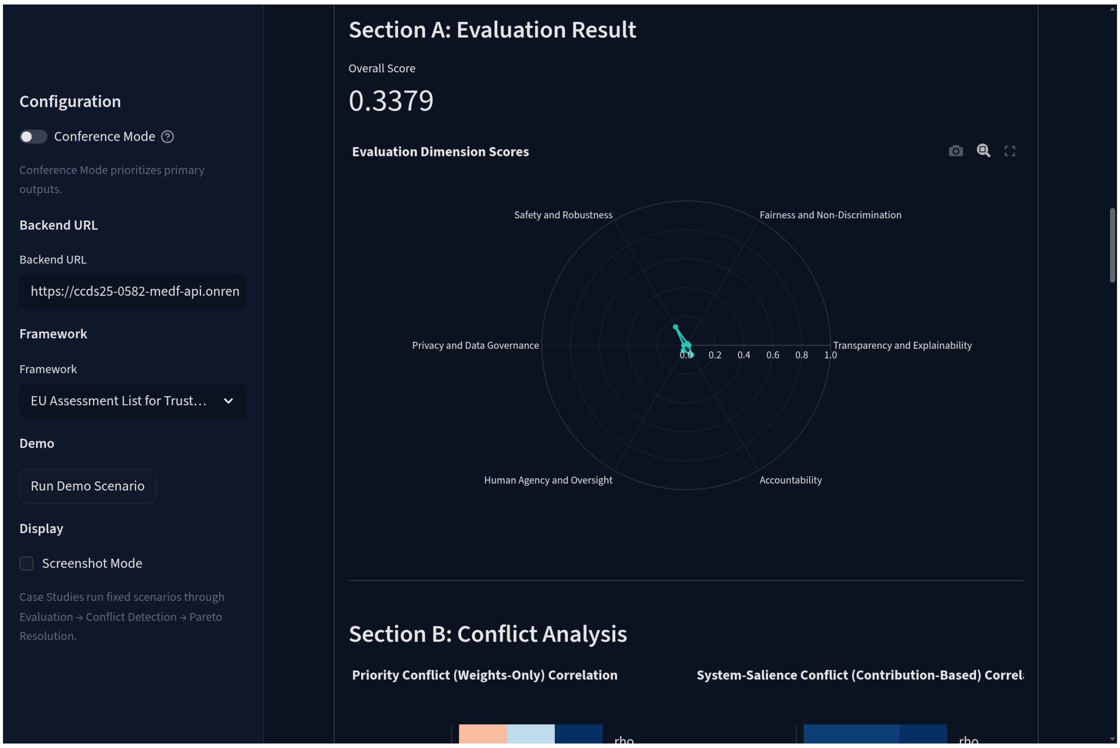

Figure 10.6: Case Study 1 (Facial Recognition) Results: overall score of 0.3379 and per-stakeholder breakdown.

By providing these interactive visualizations, the Streamlit dashboard makes the complex, multi-dimensional results of the MEDF evaluation accessible and interpretable for a non-technical audience, facilitating a more informed and data-driven discussion about AI ethics.

Chapter 11

Case Study Evaluation

To validate the practical utility and analytical capabilities of the MEDF platform, three real-world case studies were selected for evaluation. These cases were chosen to represent a diverse set of AI applications, deployment contexts, and ethical challenges. For each case, baseline dimension scores were derived from publicly available documentation, investigative reports, and academic analyses. The default stakeholder profiles (Developer, Regulator, Affected Community) were used to simulate a multi-stakeholder evaluation. This chapter presents the setup, scoring methodology, and detailed results for each case study.

11.1 Stakeholder Weight Configuration

Before presenting the individual case studies, it is important to define the stakeholder weight profiles used across all three evaluations. These weights represent the relative priority each stakeholder group assigns to each of the six Unified Dimensions. The default profiles, as defined in the MEDF repository and verified against the source code in `app/framework_registry.py`, are presented in Table 11.1.

Table 11.1: Default Stakeholder Weight Profiles: developer, regulator, and affected community priorities.

Dimension	Developer	Regulator	Affected Community
Transparency	0.10	0.20	0.10
Fairness	0.15	0.20	0.30
Safety	0.30	0.10	0.10
Privacy	0.15	0.15	0.15
Human Oversight	0.15	0.10	0.20
Accountability	0.15	0.25	0.15
Total	1.00	1.00	1.00

These profiles reflect reasonable assumptions about stakeholder priorities. The Developer profile emphasizes Safety (0.30), reflecting a focus on building reliable systems. The Regulator profile distributes weight with emphasis on Accountability (0.25), Transparency (0.20), and Fairness (0.20), reflecting a compliance-oriented perspective. The

Affected Community profile places the highest weight on Fairness (0.30) and Human Oversight (0.20), reflecting the concerns of individuals who are directly impacted by the AI system's decisions.

11.1.1 Note on Conflict Analysis Methodology

The MEDF conflict analysis engine computes Spearman's rank correlation between stakeholder contribution vectors, which are the element-wise products of each stakeholder's effective weights and the system's normalized dimension scores. Because the contribution vectors depend on the system's dimension scores, the conflict matrices differ across case studies even when the same default stakeholder weights are used. This is the **system-salience (contribution-based)** conflict metric described in Chapter 8. All conflict matrices presented in this chapter use this contribution-based method, as it provides case-specific insights into where stakeholders actually agree or disagree given the particular ethical profile of each system.

11.2 Case Study 1: Metropolitan Police Live Facial Recognition

11.2.1 System Description

This case study examines the use of live facial recognition (LFR) technology by the Metropolitan Police Service in London. The system involves deploying cameras in public spaces to scan faces in real-time and compare them against a watchlist of individuals wanted by the police [35]. The deployment has been highly controversial, raising significant concerns about privacy, mass surveillance, and algorithmic bias [36]. Independent evaluations have found that the system exhibits higher error rates for women and people of color [7], raising serious questions about its fairness. Civil liberties organizations have argued that the mere presence of such technology in public spaces creates a chilling effect on the right to protest and freedom of assembly.

11.2.2 Stakeholder Identification

Three primary stakeholder groups were identified for this evaluation:

- **Developer/Operator (Metropolitan Police).** Primarily concerned with public safety, the operational effectiveness of the system, and its robustness in identifying wanted individuals.

- **Regulator (UK Information Commissioner's Office).** Focused on legal compliance, particularly with data protection law (UK GDPR), transparency of the system's operation, and the accountability of the deploying organization.
- **Affected Community (General Public).** Concerned about the impact on their privacy, the risk of being misidentified (fairness), and the lack of consent for being subjected to biometric surveillance.

11.2.3 Baseline Dimension Scores

The baseline dimension scores for the LFR system were set to reflect its known ethical challenges, as documented in reports by organizations like Big Brother Watch and the findings of the UK Information Commissioner's Office [36]. The scores, presented in Table 11.2, reflect very low performance on privacy and fairness, and moderate performance on safety and accountability.

Table 11.2: Baseline Dimension Scores: live facial recognition system inputs.

Dimension	**Score (1–7)**	**Normalized (0–1)**
Transparency and Explainability	2.5	0.2500
Fairness and Non-discrimination	1.5	0.0833
Safety and Robustness	5.0	0.6667
Privacy and Data Governance	1.5	0.0833
Human Agency and Oversight	2.5	0.2500
Accountability	4.0	0.5000

11.2.4 MEDF Evaluation Results

The MEDF scoring engine was run using the TOPSIS method with all three governance frameworks (ALTAI, NIST RMF, MGAF). The per-stakeholder and per-framework results are summarized in Table 11.3.

Table 11.3: Evaluation Results: live facial recognition per-stakeholder scores.

Framework	**Overall**	**Developer**	**Regulator**	**Affected Community**
EU ALTAI	0.3379	0.4583	0.3092	0.2462
NIST AI RMF	0.4637	0.5552	0.4537	0.3823
Singapore MGAF	0.3582	0.4344	0.3497	0.2905
All Frameworks	0.3866	0.4826	0.3709	0.3063

The overall aggregated score across all stakeholders and frameworks was approximately 0.39, corresponding to a **Critical Risk** classification. The results clearly show a significant divergence between the Developer perspective (average score of 0.48, driven by the high Safety score) and the Affected Community perspective (average score of 0.31, driven by the very low Fairness and Privacy scores). The Regulator's score falls in between, reflecting a more balanced weighting.

11.2.5 Conflict Analysis

The contribution-based Spearman rank correlation analysis between stakeholder pairs produced the results shown in Table 11.4. These values reflect how the stakeholders' priorities interact with the specific ethical profile of the facial recognition system.

Table 11.4: Stakeholder Conflict Matrix: live facial recognition contribution-based Spearman ρ.

	Developer	**Regulator**	**Affected Community**
Developer	1.00	0.89	0.94
Regulator	0.89	1.00	0.94
Affected Community	0.94	0.94	1.00

The high positive correlations ($\rho = 0.89$ to 0.94) indicate **low conflict** across all stakeholder pairs for this case. This result may appear counterintuitive given the divergent evaluation scores, but it reflects that the contribution-based analysis measures agreement on the *ranking* of dimension contributions, not on the absolute scores. The system's extreme ethical profile, with very low Fairness and Privacy scores and a high Safety score, creates a clear ranking that all stakeholders tend to agree on, even though they weight the dimensions differently. The primary conflicting dimensions are Accountability vs. Safety (Developer-Regulator) and Accountability vs. Safety (Developer-Affected Community). The Pareto optimization engine proposed consensus weight vectors that typically involved increasing the weight on Fairness and Privacy at the expense of Safety, suggesting that any acceptable path forward would require significant improvements in these areas.

11.3 Case Study 2: iTutorGroup Automated Hiring Screening

11.3.1 System Description

This case involves an automated hiring system used by the company iTutorGroup, which was the subject of an investigation by the U.S. Equal Employment Opportunity Commission (EEOC). The EEOC found that the company had programmed its software to automatically reject applicants over the age of 55 for tutoring positions, constituting systemic age discrimination [9]. The case resulted in a settlement of $365,000 and highlighted the risks of embedding discriminatory rules directly into automated decision-making systems.

11.3.2 Stakeholder Identification

- **Developer (iTutorGroup Engineering Team).** Focused on building an efficient screening system that meets the company's operational requirements.
- **Regulator (EEOC).** Focused on ensuring compliance with anti-discrimination laws, specifically the Age Discrimination in Employment Act (ADEA).
- **Affected Community (Job Applicants).** Concerned about being treated fairly in the hiring process, regardless of age or other protected characteristics.

11.3.3 Baseline Dimension Scores

The baseline scores for this case were set to reflect the system's primary ethical failure: a deliberate and systemic lack of fairness. The system was assigned the lowest possible score on the Fairness dimension. Other dimensions received moderate scores, as the system was functional from a purely technical standpoint.

Table 11.5: Baseline Dimension Scores: iTutorGroup hiring algorithm system inputs.

Dimension	**Score (1–7)**	**Normalized (0–1)**
Transparency and Explainability	2.5	0.2500
Fairness and Non-discrimination	1.5	0.0833
Safety and Robustness	3.0	0.3333
Privacy and Data Governance	3.0	0.3333
Human Agency and Oversight	2.0	0.1667
Accountability	4.5	0.5833

11.3.4 MEDF Evaluation Results

The evaluation results, presented in Table 11.6, show universally low scores across all stakeholders and frameworks.

Table 11.6: Evaluation Results: iTutorGroup hiring algorithm per-stakeholder scores.

Framework	Overall	Developer	Regulator	Affected Community
EU ALTAI	0.3022	0.3185	0.3418	0.2462
NIST AI RMF	0.4155	0.3787	0.4892	0.3785
Singapore MGAF	0.3254	0.3292	0.3702	0.2766
All Frameworks	0.3477	0.3421	0.4004	0.3005

The overall aggregated score was approximately 0.35, corresponding to a **Critical Risk** classification. This case study demonstrates MEDF's ability to correctly identify and penalize systems with a single, catastrophic ethical flaw. Even though the system had a moderate Accountability score (reflecting that the company was eventually held accountable), the failure on Fairness was so severe that it rendered the complete system unacceptable from every stakeholder's perspective.

11.3.5 Conflict Analysis

The contribution-based conflict analysis for this case revealed moderate levels of stakeholder disagreement, as shown in Table 11.7.

Table 11.7: Stakeholder Conflict Matrix: iTutorGroup contribution-based Spearman ρ.

	Developer	Regulator	Affected Community
Developer	1.00	0.54	0.77
Regulator	0.54	1.00	0.54
Affected Community	0.77	0.54	1.00

The Developer-Regulator and Regulator-Affected Community pairs show **moderate** conflict ($\rho = 0.54$), while the Developer-Affected Community pair shows **low** conflict ($\rho = 0.77$). The primary conflicting dimensions for the Developer-Regulator pair are Safety and Transparency, reflecting their different priorities when the system's ethical profile includes moderate Safety but low Transparency scores. This case demonstrates how the contribution-based conflict metric captures case-specific dynamics: unlike the facial recognition case (where all pairs showed low conflict), the hiring algorithm's more ambiguous ethical profile produces more differentiated conflict levels.

11.4 Case Study 3: Royal Free NHS and DeepMind Streams App

11.4.1 System Description

This case study examines the deployment of the "Streams" mobile application, developed by Google's DeepMind, at the Royal Free London NHS Foundation Trust. The app was designed to help clinicians detect acute kidney injury (AKI) more quickly by processing patient data and sending alerts to medical staff. The project became controversial when it was revealed that DeepMind had been given access to the identifiable health records of approximately 1.6 million patients without their explicit consent, leading to a ruling by the UK's Information Commissioner's Office that the hospital had breached data protection law [37]. This case presents a complex ethical trade-off: the system had a clear and potentially life-saving clinical benefit, but it was achieved through a significant violation of patient privacy.

11.4.2 Stakeholder Identification

- **Developer (DeepMind/Google).** Focused on the clinical utility and safety of the application, demonstrating that AI can improve healthcare outcomes.
- **Regulator (UK ICO and NHS Data Guardian).** Focused on ensuring that the processing of sensitive health data complied with data protection law and that patients' rights were respected.
- **Affected Community (NHS Patients).** Concerned about the unauthorized use of their personal health data, but also potentially benefiting from the improved clinical care the system could provide.

11.4.3 Baseline Dimension Scores

The baseline scores for the Streams app reflect a different ethical profile from the previous two cases. The system was designed with a clear clinical benefit, leading to high scores on Safety and Human Oversight (as it was a decision-support tool that augmented, rather than replaced, clinician judgment). However, its major failure was in Privacy and Data Governance.

Table 11.8: Baseline Dimension Scores: NHS-DeepMind Streams system inputs.

Dimension	Score (1–7)	Normalized (0–1)
Transparency and Explainability	4.0	0.5000
Fairness and Non-discrimination	4.0	0.5000
Safety and Robustness	6.0	0.8333
Privacy and Data Governance	3.5	0.4167
Human Agency and Oversight	5.0	0.6667
Accountability	4.5	0.5833

11.4.4 MEDF Evaluation Results

The MEDF evaluation for the Streams app produced a more nuanced result than the previous two cases, as shown in Table 11.9.

Table 11.9: Evaluation Results: NHS-DeepMind Streams per-stakeholder scores.

Framework	Overall	Developer	Regulator	Affected Community
EU ALTAI	0.5841	0.6673	0.5376	0.5475
NIST AI RMF	0.6258	0.7092	0.5836	0.5847
Singapore MGAF	0.5910	0.6479	0.5470	0.5780
All Frameworks	0.6003	0.6748	0.5561	0.5701

The overall aggregated score was approximately 0.60, corresponding to a **Medium Risk** classification. This result is significantly higher than the previous two cases, reflecting the system's genuine strengths in Safety and Human Oversight. However, the score is still well below the "Low Risk" threshold, pulled down by the Privacy failure. The divergence between the Developer score (0.68) and the Affected Community score (0.57) is notable and reflects the core ethical tension in this case.

11.4.5 Conflict Analysis

The contribution-based conflict analysis for this case revealed the highest levels of stakeholder disagreement among the three case studies, as shown in Table 11.10.

Table 11.10: Stakeholder Conflict Matrix: NHS-DeepMind Streams contribution-based Spearman ρ.

	Developer	**Regulator**	**Affected Community**
Developer	1.00	−0.14	0.49
Regulator	−0.14	1.00	0.03
Affected Community	0.49	0.03	1.00

The Developer-Regulator pair shows **high conflict** ($\rho = -0.14$), and the Regulator-Affected Community pair also shows **high conflict** ($\rho = 0.03$). The Developer-Affected Community pair shows **moderate conflict** ($\rho = 0.49$). The primary conflicting dimensions are Transparency and Human Oversight, reflecting the core tension in this case: the Developer values the system's clinical utility (Safety) while the Regulator prioritizes the data governance failure (Privacy, Transparency). This case study demonstrates the full power of the contribution-based conflict metric. Unlike the facial recognition case where the extreme ethical profile led to broad agreement, the healthcare case's more balanced profile exposes genuine disagreements about which dimensions matter most.

The Pareto optimization engine generated a set of consensus solutions that explored the trade-off between Safety/Human Oversight and Privacy/Transparency, providing a quantitative basis for the kind of nuanced policy discussion that this case demands.

11.5 Cross-Case Comparative Analysis

A summary comparison of the three case studies is presented in Table 11.11. This comparison highlights the diversity of the ethical profiles and the ability of MEDF to differentiate between them.

Table 11.11: Cross-Case Comparison: evaluation results across all three case studies.

Case Study	**Overall**	**Risk**	**Weakness**	**Min Pairwise ρ**	**Conflict**
Facial Recognition	0.3866	Critical	Fairness, Privacy	0.89	Low
iTutorGroup Hiring	0.3477	Critical	Fairness	0.54	Moderate
NHS-DeepMind Streams	0.6003	Medium	Privacy	−0.14	High

The cross-case analysis reveals several important patterns. First, the LFR and iTutorGroup cases, which involve clear and severe ethical violations, both received low scores and critical risk classifications. This validates the framework's ability to correctly flag problematic systems. Second, the NHS-DeepMind case, which involves a more complex trade-off between benefit and harm, received a higher score but still fell within the Medium risk range, demonstrating the framework's sensitivity to nuance.

Thirdly, and most importantly, the contribution-based conflict analysis produces *different* results across cases, even though the same default stakeholder weights are used. The facial recognition case shows low conflict (all stakeholders agree on the ranking of dimension contributions because the ethical profile is extreme), the hiring case shows moderate conflict, and the healthcare case shows high conflict (the more balanced ethical profile exposes genuine disagreements about priorities). This validates MEDF's ability to distinguish between genuinely contentious cases and cases with clear ethical violations through the contribution-based conflict analysis approach.

Note that the "Overall" column in each evaluation table represents the aggregated TOPSIS closeness coefficient across all three stakeholders for a given framework, while the "All Frameworks" row represents the aggregated score when all three frameworks are evaluated simultaneously. The cross-case comparison table uses the all-frameworks overall score for each case study.

These results demonstrate that the MEDF platform is a versatile and informative tool for ethical AI evaluation. It can handle a range of ethical profiles, from clear-cut violations to complex trade-offs, and it provides both quantitative scores and qualitative insights (through conflict analysis) that can inform real-world governance decisions.

Chapter 12

Discussion

The development and evaluation of the MEDF platform have yielded several important insights into the challenges and opportunities of computational AI ethics. This chapter discusses the critical findings from the project, the implications of the MEDF approach, and the role of such a tool in the broader AI governance ecosystem.

12.1 Critical Findings

12.1.1 The Value of Quantification

One of the most significant findings from the case study evaluations is the power of quantification in clarifying ethical debates. Abstract discussions about whether a system is "fair" or "transparent" can be unproductive. By requiring assessors to assign a numerical score (even a subjective one) and forcing stakeholders to articulate their priorities as numerical weights, MEDF transforms a vague conversation into a structured, analytical process. The final scores, while not an absolute measure of ethicality, serve as a powerful, data-driven focal point for discussion. For example, seeing a system score 0.35 overall, with a High Risk flag on fairness, is a much more concrete and actionable starting point than simply stating that the system has "fairness issues."

12.1.2 Surfacing Implicit Trade-offs

Ethical decision-making is often about managing trade-offs. The MEDF framework makes these trade-offs explicit. The conflict analysis between the Developer and Affected Community in the facial recognition case study, for instance, does not just say they disagree. It shows *why* they disagree by pinpointing the dimensions (privacy vs. safety) where their values diverge. Furthermore, the Pareto optimization engine demonstrates that there is often no single "correct" answer. Instead, there is a frontier of viable compromises. By presenting this frontier to decision-makers, MEDF shifts the conversation from a confrontational, zero-sum argument to a collaborative exploration of possible solutions. It provides a map of the solution space, allowing stakeholders to negotiate more effectively.

12.1.3 The Importance of a Multi-Framework Approach

Integrating three distinct governance frameworks (ALTAI, NIST RMF, MGAF) proved to be a valuable design decision. The case study evaluations showed that a system's ethical score varies depending on the framework used for evaluation. This reflects the reality that different regulatory regimes have different priorities. A system that is compliant with the industry-focused MGAF might not satisfy the more stringent, rights-based requirements of ALTAI. MEDF's ability to evaluate against multiple frameworks simultaneously provides a more comprehensive and globally-aware assessment, which is crucial for organizations deploying AI systems in multiple jurisdictions.

12.2 Implications and Role in AI Governance

The MEDF platform is not intended to be an automated "ethics oracle" that delivers a final, unquestionable verdict. Rather, it is a **decision-support tool**. Its primary role is to structure and inform human judgment, not to replace it. The scores and analyses it produces are best understood as inputs into a broader governance process that includes human deliberation, legal review, and public consultation.

MEDF can be integrated into the AI development lifecycle at several critical points:

- **Design Phase.** As a complement to qualitative methods like the FID framework [1], MEDF can be used to model the potential ethical impacts of different design choices before implementation begins.
- **Internal Audit and Review.** Development teams can use MEDF to conduct internal ethical audits of their systems, identify areas of high risk, and document their due diligence.
- **External Reporting and Compliance.** The outputs from MEDF can be used to generate reports for regulators, auditors, and the public, providing transparent and data-driven evidence of a system's ethical posture and the process used to evaluate it.
- **Procurement.** Organizations looking to procure third-party AI systems could use MEDF to evaluate and compare the ethical performance of different vendors in a standardized way.

By providing a common language and a shared analytical framework, MEDF can help to foster a more productive and evidence-based dialogue between the diverse actors in the AI ecosystem, from the engineers building the systems to the regulators overseeing them and the public living with their consequences.

12.3 Comparison with Existing Approaches

It is useful to position MEDF relative to other approaches in the field. A systematic review by Morley et al. [38] categorized publicly available AI ethics tools into five types and found that most remain at the principles or guidelines level, with few providing computational or decision-support functionality. Existing tools for ethical AI assessment can be broadly categorized into three types: (1) qualitative checklists and guidelines, such as the raw ALTAI checklist itself, (2) technical fairness toolkits, such as IBM's AI Fairness 360 or Google's What-If Tool, which focus on detecting statistical bias in model outputs, and (3) governance platforms, which provide organizational workflows for managing AI risk.

MEDF occupies a distinct position in this landscape. Unlike qualitative checklists, it provides a quantitative, computational output. Unlike technical fairness toolkits, it operates at a higher level of abstraction, evaluating the system against a broad set of ethical dimensions rather than focusing solely on statistical bias. And unlike most governance platforms, it explicitly models multi-stakeholder conflict and provides algorithmic tools for consensus generation. The Z-Inspection process [39] shares some conceptual goals with MEDF in assessing trustworthy AI, but it relies on qualitative expert panels rather than computational scoring and optimization. This combination of features makes MEDF a unique contribution to the field, filling a gap between high-level principles and low-level technical metrics.

12.4 Future Work

Several promising directions for future research and development emerge from this project:

- **Empirical Validation.** Conducting user studies with AI practitioners, ethicists, and policymakers to evaluate the usability and perceived value of the MEDF platform. This would provide empirical evidence to complement the technical validation presented in this report.
- **Automated Score Generation.** Exploring the use of Natural Language Processing (NLP) techniques to automatically generate baseline dimension scores from system documentation, audit reports, or news articles. This would reduce the reliance on subjective manual scoring.
- **Dynamic and Longitudinal Evaluation.** Extending MEDF to support longitudinal tracking of a system's ethical posture over time, allowing organizations to monitor how their ethical performance evolves as the system is updated and the

regulatory landscape changes.

- **Integration with CI/CD Pipelines.** Developing plugins or integrations that allow MEDF evaluations to be triggered automatically as part of a Continuous Integration/Continuous Deployment (CI/CD) pipeline, embedding ethical checks directly into the software development workflow.
- **Expanded Framework Library.** Adding support for additional governance frameworks, such as the IEEE Ethically Aligned Design standards, the OECD AI Principles, or sector-specific regulations for healthcare, finance, and criminal justice.

Chapter 13

Limitations

While the MEDF platform represents a significant step towards operationalizing AI ethics, it is essential to acknowledge its limitations. These limitations define the boundaries of the current implementation and highlight important areas for future research and development. The critical limitations can be grouped into three categories: subjectivity of inputs, simplification of models, and scope of evaluation.

13.1 Subjectivity of Inputs

The adage “garbage in, garbage out” applies forcefully to the MEDF framework. The validity of the platform’s output is fundamentally dependent on the quality and integrity of its inputs.

- **Baseline Score Subjectivity.** The initial dimension scores for an AI system are a critical input. In the current implementation, these scores are assumed to be provided by the user (e.g., the development team). These scores are inherently subjective and can be influenced by the assessor’s biases, knowledge, and incentives. A team might be motivated to provide inflated scores to make their system appear more ethical than it is. While MEDF provides a framework for analyzing these scores, it does not have a built-in mechanism for independently verifying their accuracy against ground truth.
- **Stakeholder Weight Subjectivity.** Similarly, the stakeholder weights are also subjective. While they are intended to represent the genuine priorities of different groups, determining the “correct” weights for a group like the “Affected Community” is a significant challenge in itself. The default profiles provided in MEDF are based on reasonable assumptions, but they are not the result of empirical sociological research. A comprehensive and truly representative evaluation would require a more rigorous process for eliciting these weights from the actual stakeholder groups.

13.2 Simplification of Models

To create a tractable computational framework, MEDF necessarily simplifies the complex reality of ethical decision-making.

- **Linearity and Independence Assumptions.** The scoring models (TOPSIS and WSM) and the weighting schemes implicitly assume a degree of independence between the ethical dimensions. In reality, these dimensions are often deeply intertwined. For example, improving transparency might have a positive impact on accountability but could have a negative impact on safety if it exposes system vulnerabilities. The current models do not explicitly capture these complex, non-linear interactions.
- **Reduction of Ethics to a Score.** The act of reducing a complex ethical profile to a single numerical score is a powerful simplification, but it is also a lossy one. A single score cannot capture the full nuance of an ethical dilemma. Two systems could receive the same overall score but have vastly different ethical profiles: one might be moderately weak across all dimensions, while the other might excel in five dimensions but have a critical flaw in one. While MEDF provides dimensional breakdowns to mitigate this, the focus on a single score can risk oversimplification. The use of a seven-point Likert scale further constrains the resolution of the input signal, meaning that two systems with meaningfully different ethical profiles could receive identical baseline scores.

13.3 Scope of Evaluation

The current implementation of MEDF has a defined scope, and several important aspects of AI governance are not yet included.

- **No Primary Research.** The project mandate explicitly excluded primary research such as user surveys, interviews, or statistical user studies. As a result, the framework's usability and the validity of its default assumptions have not been empirically tested with real-world users.
- **Focus on Post-Hoc Evaluation.** MEDF is primarily designed for the evaluation of existing systems or well-defined designs. While it can be used to inform the design process, it is not a generative tool that actively assists in the creation of more ethical designs from scratch. It is an evaluative, not a creative, framework.
- **Limited Harm Taxonomy.** The harm assessment module provides a high-level estimation of risk. A more sophisticated implementation would incorporate a more detailed and evidence-based taxonomy of AI-related harms, potentially drawing on fields like law and social science to create a more granular risk model.
- **Incomplete Separation of Concerns.** The conflict analysis logic is currently implemented within the API router (`routers/conflicts.py`) rather than an extracted standalone engine module. While the functionality is complete and all tests pass, this architectural shortcut represents technical debt. A future refactoring

should separate the conflict analysis logic into a dedicated module to complete the intended separation of concerns.

These limitations do not invalidate the MEDF approach. Rather, they clarify its role as a specific type of decision-support tool and provide a clear roadmap for future work aimed at building more comprehensive, empirically-grounded, and nuanced tools for computational ethics.

Chapter 14

Conclusion

This project set out to address a critical challenge in the field of Artificial Intelligence: the gap between the articulation of ethical principles and their practical, rigorous implementation. The result of this effort is the Multi-stakeholder Ethical Decision Framework (MEDF), a comprehensive, open-source software platform for the systematic evaluation of AI systems against established governance standards. By successfully designing, building, and testing this framework, the project makes several important contributions to the field of AI ethics and governance.

Firstly, MEDF demonstrates that it is possible to operationalize high-level ethical principles into a quantitative, computational model. By harmonizing leading governance frameworks into a unified six-dimension structure and employing established multi-criteria decision-making algorithms, the platform provides a methodology for moving beyond subjective debate to a more structured and data-driven form of ethical analysis. This provides a valuable tool for developers, auditors, and regulators who require a consistent and defensible way to measure and compare the ethical performance of AI systems.

Secondly, the framework places the management of stakeholder conflict at the center of the ethical evaluation process. By incorporating configurable stakeholder weightings and a dedicated conflict resolution engine, MEDF explicitly acknowledges that ethical decision-making is not about finding a single, objectively "correct" answer, but about navigating the complex and often competing values of diverse groups. The use of Pareto optimization to generate a menu of viable compromises provides a practical mechanism for facilitating more productive and transparent negotiations around ethical trade-offs.

Thirdly, the project delivers a functional and reproducible research artifact. The entire platform, including the source code, documentation, test suite, and case study data, is provided as an open-source project. This commitment to transparency and reproducibility ensures that the findings of this report can be independently verified and, more importantly, that the MEDF platform can serve as a foundation for future research and development in the growing field of computational ethics.

While the limitations of the current implementation, such as the subjectivity of inputs and the simplification of complex ethical models, are acknowledged, the MEDF platform successfully achieves its primary objective. It provides an extensible and practical tool for bridging the principles-to-practice gap in AI ethics, with identified technical debt such as router-level concentration of conflict-analysis logic providing a clear roadmap for continued development. Future work can build on this foundation by incorporating more sophisticated risk taxonomies, conducting empirical user studies to validate the framework's usability, and exploring the integration of MEDF with other stages of the AI development lifecycle.

In conclusion, the responsible development and deployment of AI is one of the most pressing challenges of our time. It requires not only a commitment to ethical principles but also the creation of practical tools and methodologies to put those principles into action. The MEDF project represents a tangible contribution to this effort, providing a clear and powerful demonstration of how computational techniques can be harnessed to foster a more rigorous, transparent, and inclusive approach to AI governance.

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Appendix A

API References

This appendix provides a summary of the MEDF REST API endpoints. The complete, interactive API documentation is available at `/api/openapi.json` when the FastAPI server is running.

A.1 Evaluation Endpoints

Table A.1: MEDF API Endpoint References.

Method	Endpoint	Description
`GET`	`/api/health`	Returns the health status of the API server.
`GET`	`/api/frameworks`	Lists all available governance frameworks and their definitions.
`GET`	`/api/stakeholders`	Lists all registered stakeholder profiles.
`POST`	`/api/stakeholders`	Creates a new stakeholder profile with custom weights.
`POST`	`/api/evaluate`	Runs an ethical evaluation given an AI system context, framework IDs, stakeholder IDs, and weight overrides. Returns per-framework and aggregated scores.
`POST`	`/api/conflicts`	Computes the stakeholder conflict matrix (contribution-based Spearman ρ) for a given evaluation context. Requires an `ai_system` object.
`POST`	`/api/pareto`	Runs NSGA-II optimization to find Pareto-optimal consensus weight vectors.

A.2 Request Schema

The primary request schemas are defined as Pydantic models in `app/models.py`. The critical fields are:

- `ai_system`: A nested object containing `id` (string), `name` (string), `description` (optional string), and `context` (object with `dimension_scores`).

- `framework_ids`: A list of framework identifier strings (e.g., `["eu_altai"]`).
- `stakeholder_ids`: A list of stakeholder identifier strings (e.g., `["developer"]`).
- `weights`: A nested dictionary mapping each stakeholder ID to its dimension weight vector.
- `scoring_method`: Either `"topsis"` or `"wsm"` (default: `"topsis"`).

Dimension keys use the full Unified Dimension identifiers:

`transparency_explainability`, `fairness_nondiscrimination`, `safety_robustness`, `privacy_data_governance`, `human_agency_oversight`, and `accountability`.

A.3 Example: Evaluation Request

Listing A.3: Example API call to the evaluate endpoint.

```
import requests

payload = {
    "ai_system": {
        "id": "demo_facerec",
        "name": "Facial Recognition LFR",
        "description": "Metropolitan Police LFR system",
        "context": {
            "dimension_scores": {
                "transparency_explainability": 2.5,
                "fairness_nondiscrimination": 1.5,
                "safety_robustness": 5.0,
                "privacy_data_governance": 1.5,
                "human_agency_oversight": 2.5,
                "accountability": 4.0
            }
        }
    },
    "framework_ids": ["eu_altai"],
    "stakeholder_ids": ["developer", "regulator",
                        "affected_community"],
    "weights": {
        "developer": {
            "transparency_explainability": 0.10,
            "fairness_nondiscrimination": 0.15,
            "safety_robustness": 0.30,
```

```
            "privacy_data_governance": 0.15,
            "human_agency_oversight": 0.15,
            "accountability": 0.15
        },
        "regulator": {
            "transparency_explainability": 0.20,
            "fairness_nondiscrimination": 0.20,
            "safety_robustness": 0.10,
            "privacy_data_governance": 0.15,
            "human_agency_oversight": 0.10,
            "accountability": 0.25
        },
        "affected_community": {
            "transparency_explainability": 0.10,
            "fairness_nondiscrimination": 0.30,
            "safety_robustness": 0.10,
            "privacy_data_governance": 0.15,
            "human_agency_oversight": 0.20,
            "accountability": 0.15
        }
    },
    "scoring_method": "topsis"
}

response = requests.post(
    "http://localhost:8000/api/evaluate",
    json=payload
)
result = response.json()
print(f"Overall Score: {result['overall_score']:.4f}")
```

Appendix B

Framework YAML Definitions

This appendix reproduces excerpts from the YAML configuration files that define the three governance frameworks integrated into MEDF. These files are located in the `app/frameworks/` directory of the repository. The excerpts below are verified against the actual source files as of the current repository version.

B.1 EU ALTAI (`eu_altai.yaml`)

The EU ALTAI framework defines six criteria mapped to the Unified Dimensions, with a total of 22 requirements and assessment questions. The dimension weights are derived from the number of requirements per dimension divided by the total (22). Safety and Robustness and Privacy and Data Governance are tied for the highest weight (0.2273 each) due to their five requirements each.

Listing B.1: EU ALTAI Section-Count Weights: derived from requirement counts per dimension.

```yaml
id: eu_altai
name: EU Assessment List for Trustworthy AI (ALTAI)
version: "2020"
description: Baseline criteria distilled from EU ALTAI guidance.
criteria:
  - id: altai_transparency
    dimension: transparency_explainability
    requirements: [ALTAI-TR-1, ALTAI-TR-2]     # 2/22 = 0.0909
    weight: 0.0909
  - id: altai_fairness
    dimension: fairness_nondiscrimination
    requirements: [ALTAI-FR-1, ALTAI-FR-2, ALTAI-FR-3] # 3/22 = 0.1364
    weight: 0.1364
  - id: altai_safety
    dimension: safety_robustness
    requirements: [ALTAI-SF-1, ..., ALTAI-SF-5] # 5/22 = 0.2273
    weight: 0.2273
  - id: altai_privacy
    dimension: privacy_data_governance
```

```
    requirements: [ALTAI-PR-1, ..., ALTAI-PR-5] # 5/22 = 0.2273
    weight: 0.2273
  - id: altai_human_oversight
    dimension: human_agency_oversight
    requirements: [ALTAI-HO-1, ..., ALTAI-HO-4] # 4/22 = 0.1818
    weight: 0.1818
  - id: altai_accountability
    dimension: accountability
    requirements: [ALTAI-AC-1, ALTAI-AC-2, ALTAI-AC-3] # 3/22 = 0.1364
    weight: 0.1364
```

The weights sum to: $0.0909 + 0.1364 + 0.2273 + 0.2273 + 0.1818 + 0.1364 \approx 1.0001$ (negligible rounding from repeating fractions. The evaluate router normalizes weights at runtime).

B.2 NIST AI RMF (`nist_ai_rmf.yaml`)

The NIST AI Risk Management Framework defines 19 subcategories mapped to the six Unified Dimensions. The weights are derived from the number of subcategories per dimension divided by the total (19), reflecting the structural emphasis of the NIST RMF.

Listing B.2: NIST AI RMF Section-Count Weights: derived from subcategory counts per dimension.

```
id: nist_ai_rmf
name: NIST AI Risk Management Framework
version: "1.0"
# Dimension weights (from subcategory counts, 19 total):
#   transparency_explainability: 0.1053 (2/19 subcategories)
#   fairness_nondiscrimination: 0.1053 (2/19 subcategories)
#   safety_robustness:          0.2632 (5/19 subcategories)
#   privacy_data_governance:    0.1053 (2/19 subcategories)
#   human_agency_oversight:     0.1053 (2/19 subcategories)
#   accountability:             0.3158 (6/19 subcategories)
```

The weights sum to: $0.1053 \times 4 + 0.2632 + 0.3158 \approx 1.0002$ (negligible rounding from repeating fractions. The evaluate router normalizes weights at runtime).

B.3 Singapore MGAF (`sg_mgaf.yaml`)

The Singapore Model AI Governance Framework defines 20 principles mapped to the six Unified Dimensions. The weights are derived from the number of principles per dimension divided by the total (20), reflecting the framework's emphasis on transparency and accountability.

Listing B.3: Singapore MGAF Section-Count Weights: derived from principle counts per dimension.

```yaml
id: sg_mgaf
name: Singapore Model AI Governance Framework
version: "2.0"
# Dimension weights (from principle counts, 20 total):
#   transparency_explainability: 0.2500 (5/20 principles)
#   fairness_nondiscrimination: 0.1000 (2/20 principles)
#   safety_robustness:          0.1500 (3/20 principles)
#   privacy_data_governance:    0.1000 (2/20 principles)
#   human_agency_oversight:     0.2000 (4/20 principles)
#   accountability:             0.2000 (4/20 principles)
```

The weights sum to: $0.2500 + 0.1000 + 0.1500 + 0.1000 + 0.2000 + 0.2000 = 1.0000$.